Classification Evaluation

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Agenda

- Introduction
- TP | TN | FP | FN
- Accuracy
- Precision | Recall | F1
- TPR | FPR | ROC | AUC
- Confusion Matrix for Multi-Class

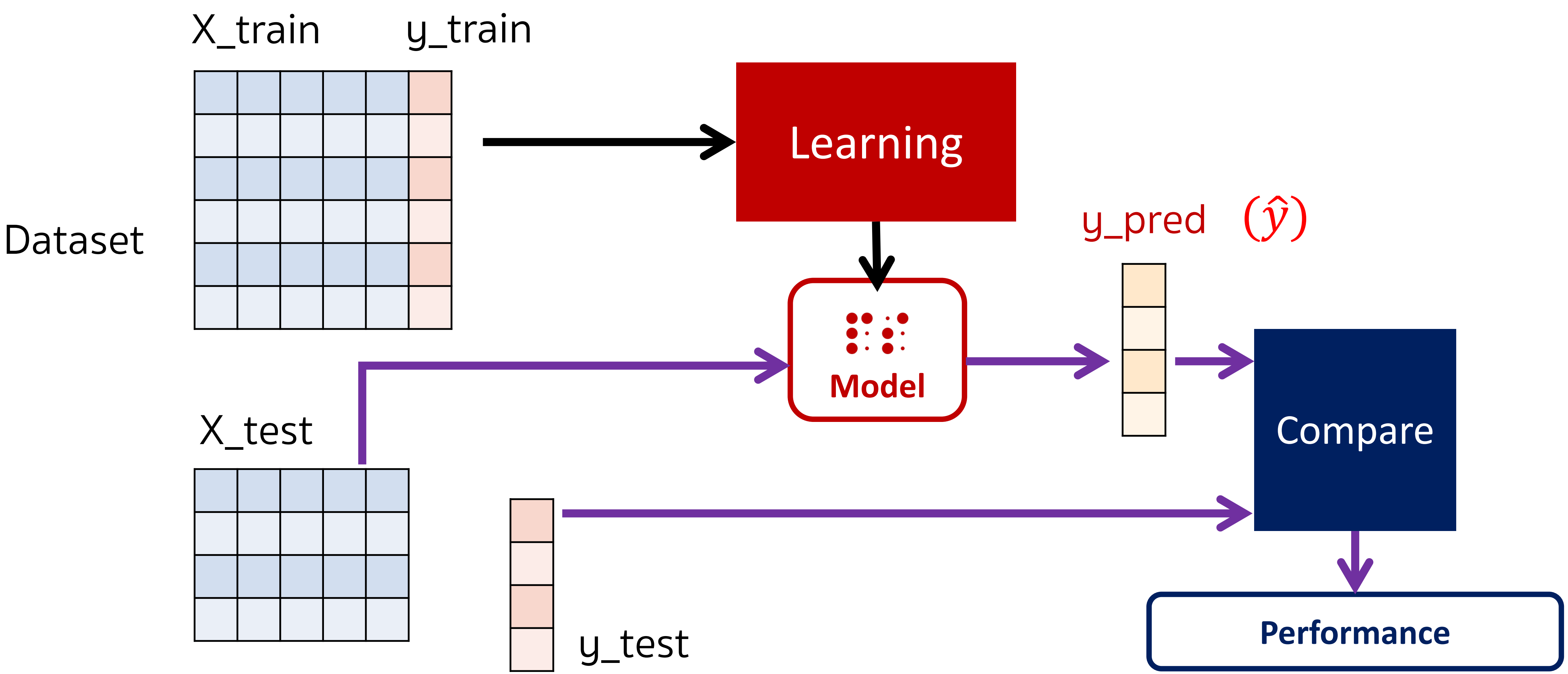
Introduction

AI Techniques

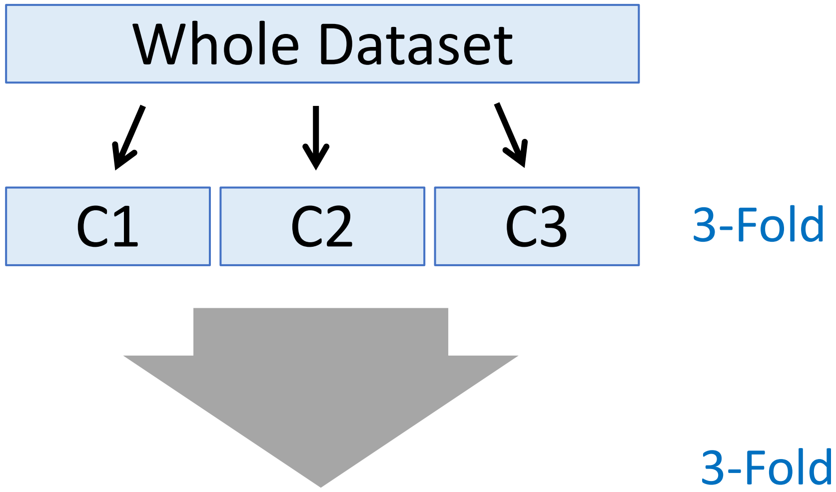
AI

- Rule-Driven
  - Expert System
  - …
- Data-Driven
  - Supervised
    - Regression
      - Linear Regression
      - Polynomial Regression
      - …
    - Classification
      - Logistic Regression
      - Naïve Bayes
      - Decision Tree
      - Neural Network
      - …
  - Unsupervised
    - K-Mean
    - DBSCAN
    - …

Train-Test Split



K-Fold Cross-Validation



3-Fold Cross-Validation

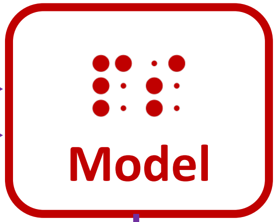
Round	Training Set			Test Set	Evaluation
1		C2	C3	C1	e1
2	C1		C3	C2	e2
3	C1	C2		C3	e3

Average Result $\mathbf{\frac{e1 + e2 + e3}{3}}$

Test Set

Sample #	has_wing	can_fly	actual_class
1	yes	yes	bird
2	yes	yes	bird
3	yes	no	 bird
4	yes	yes	 not-bird
5	no	no	not-bird
6	no	yes	not-bird
7	no	no	 bird
8	no	yes	not-bird

Classifier Did

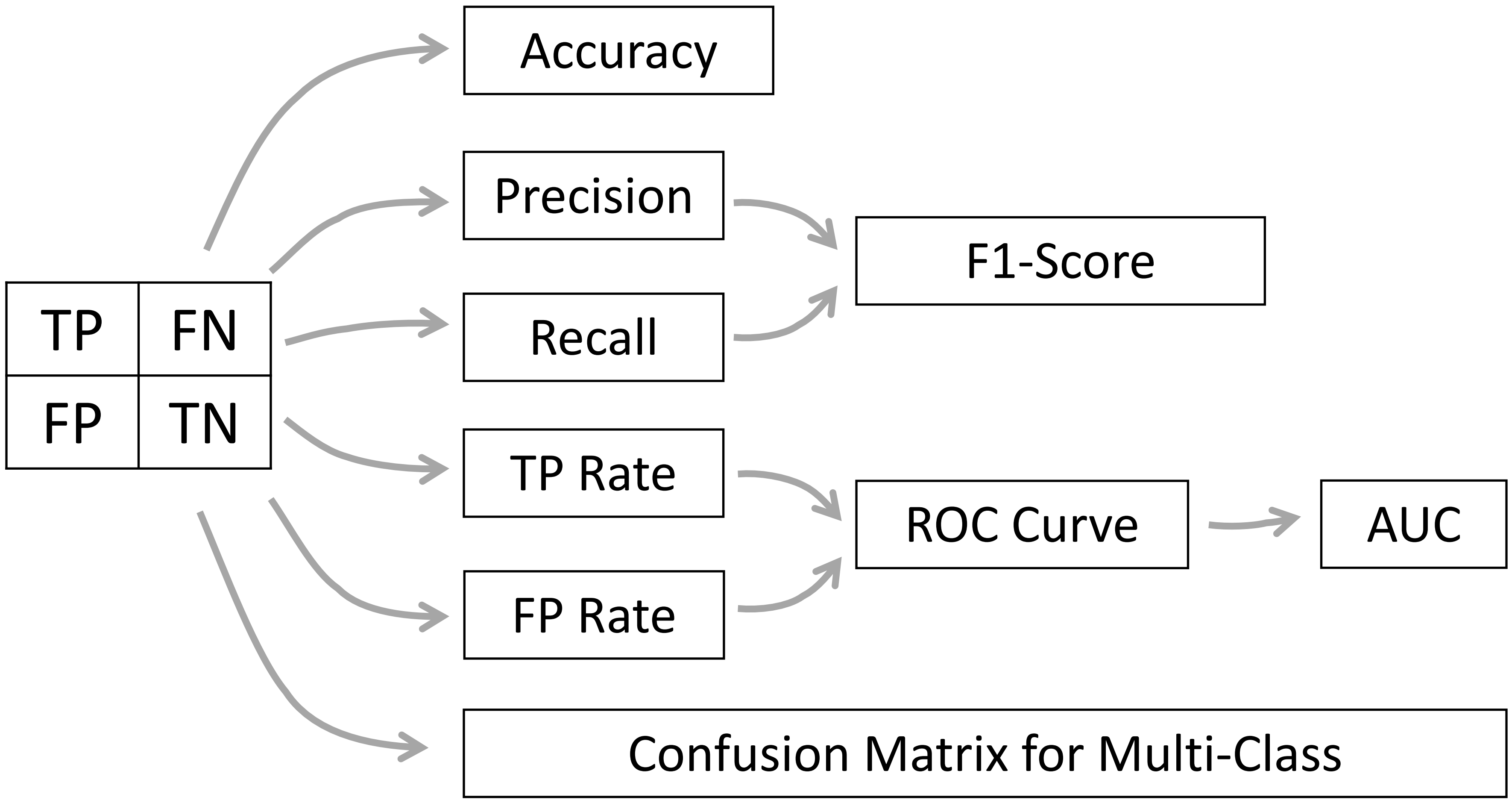


Sample #	has_wing	can_fly	actual_class	predicted
1	yes	yes	bird	bird
2	yes	yes	bird	bird
3	yes	no	 bird	not-bird
4	yes	yes	 not-bird	bird
5	no	no	not-bird	not-bird
6	no	yes	not-bird	not-bird
7	no	no	 bird	not-bird
8	no	yes	not-bird	not-bird

Correct Classification?

Sample #	has_wing	can_fly	actual_class	predicted	
1	yes	yes	bird	bird	✓
2	yes	yes	bird	bird	✓
3	yes	no	 bird	not-bird	✗
4	yes	yes	 not-bird	bird	✗
5	no	no	not-bird	not-bird	✓
6	no	yes	not-bird	not-bird	✓
7	no	no	 bird	not-bird	✗
8	no	yes	not-bird	not-bird	✓

Evaluation Methods



TP | TN | FP | FN

TP | TN | FP | FN

- There are other ways to measure different aspects of performance. In classic machine learning nomenclature, when we're dealing with binary classification, the classes are: **positive and negative**. Think of these classes in the context of disease detection:
  - **positive** – we predict the **disease is present**
  - **negative** – we predict the disease is NOT present.
- Let's now define some notations:
  - **TP** – True Positives (Samples the classifier has correctly classified as positives)
  - **TN** – True Negatives (Samples the classifier has correctly classified as negatives)
  - **FP** – False Positives (Samples the classifier has incorrectly classified as positives)
  - **FN** – False Negatives (Samples the classifier has incorrectly classified as negatives)

Ref: http://nlpforhackers.io/classification-performance-metrics/

Positive vs.. Negative

- Positive
  - It doesn't mean any Good Things
  - It means <u>a class that you want</u>
- Negative
  - It doesn't mean any Bad Things
  - It means <u>a class that you don’t want</u>

- don’t be emotional -

Ref: http://nlpforhackers.io/classification-performance-metrics/

TP | TN | FP | FN

- True-Positive
  - The model selects a class that you want.
- True-Negative
  - The model doesn’t select a class that you don’t want.
- False-Positive
  - The model selects a class that you don’t want.
- False-Negative
  - The model doesn’t select a class that you want.

Ref: http://nlpforhackers.io/classification-performance-metrics/

TP | TN | FP | FN

- True-Positive
  - The model selects a class that you want.
  - The model thinks "you want" and it thinks correctly.
- True-Negative
  - The model doesn't select a class that you don't want.
  - The model thinks "you don't want" and it thinks correctly.
- False-Positive
  - The model selects a class that you don't want.
  - The model thinks "you want" but it thinks incorrectly.
- False-Negative
  - The model doesn't select a class that you want.
  - The model thinks "you don't want" but it thinks incorrectly.

Ref: http://nlpforhackers.io/classification-performance-metrics/

TP | TN | FP | FN

	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

TP | TN | FP | FN

	predicted: **YES**	predicted: **no**
actual: **YES**	**198** *(true-positive)*	**3** *(false-negative)*
actual: **no**	**4** *(false-positive)*	**175** *(true-negative)*

TP | TN | FP | FN

Q#	Scenario	Result
1	To detect cars, the model selects a car.	
2	To detect cars, the model selects a bicycle.	
3	To detect cars, the model does not select a car.	
4	To detect cars, the model does not select a bicycle.	
5	To detect fire, the fire alarm alerts without fire.	
6	To detect thieves, the device does not alert when a thief enters into the house.	
7	To detect fire, the fire alarm does not alert under a normal situation.	
8	To detect thieves, the device alerts when a family member enters into the house.	

Ref: http://nlpforhackers.io/classification-performance-metrics/

TP | TN | FP | FN

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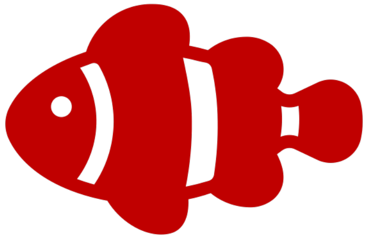
8	To detect thieves, the device alerts when a family member enters into the house.	FP

Ref: http://nlpforhackers.io/classification-performance-metrics/

Scenario

Ref: http://www.cheatbook.de/magazin/cartoons/cartoon26.htm

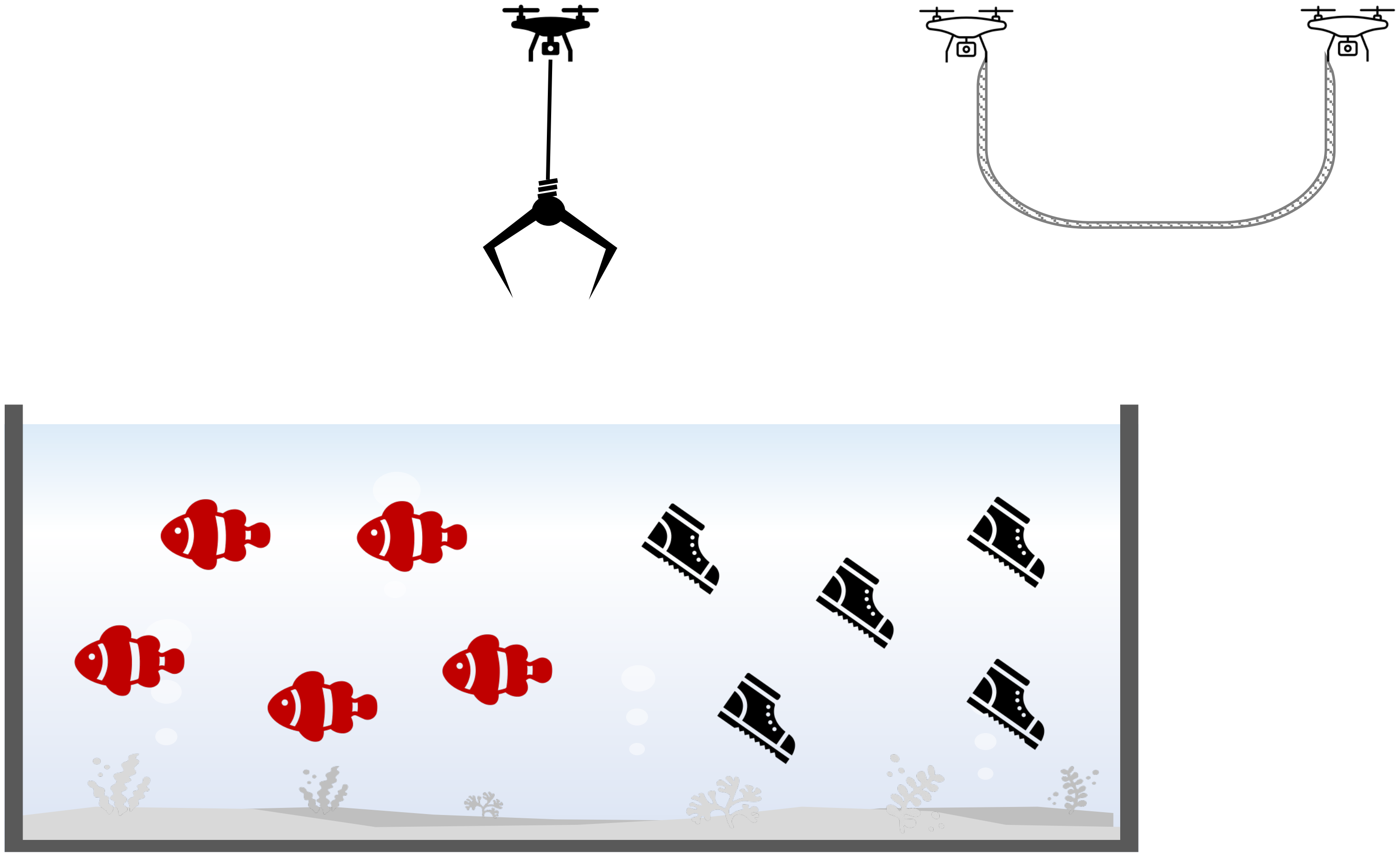
We need FISH



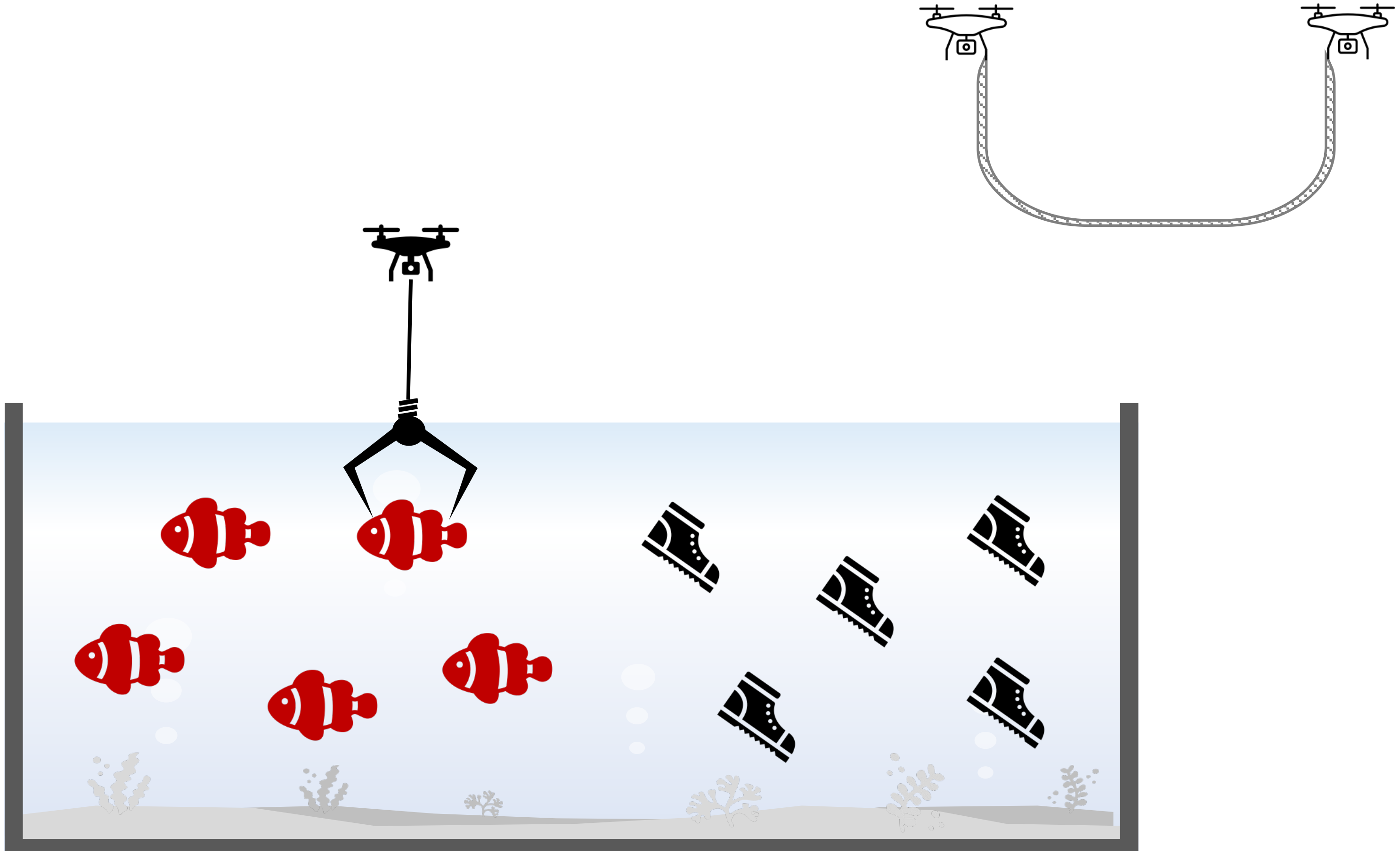
Correct Class

Incorrect Class

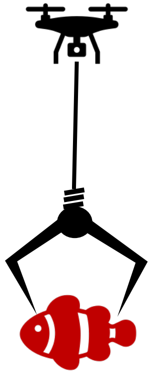
We need FISH

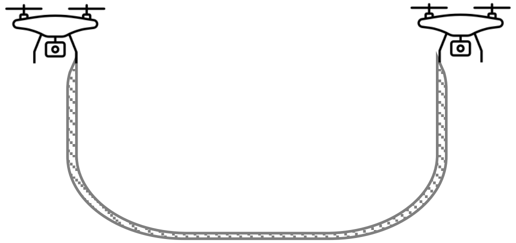


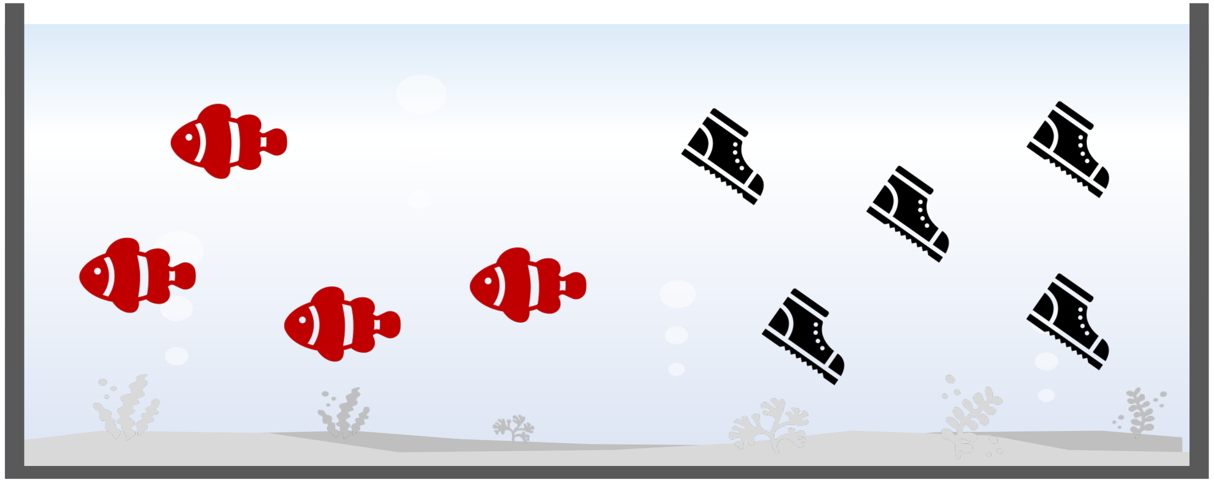
We need FISH



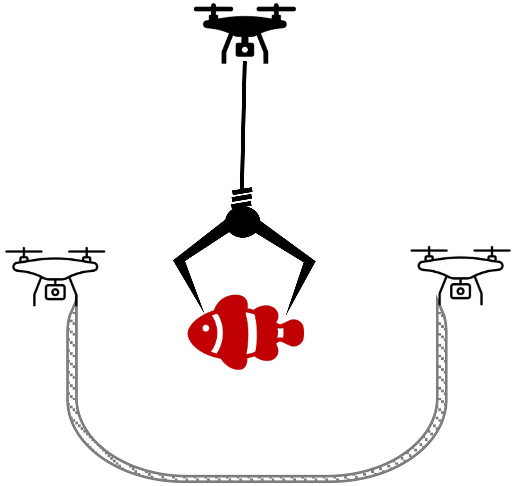
We need FISH

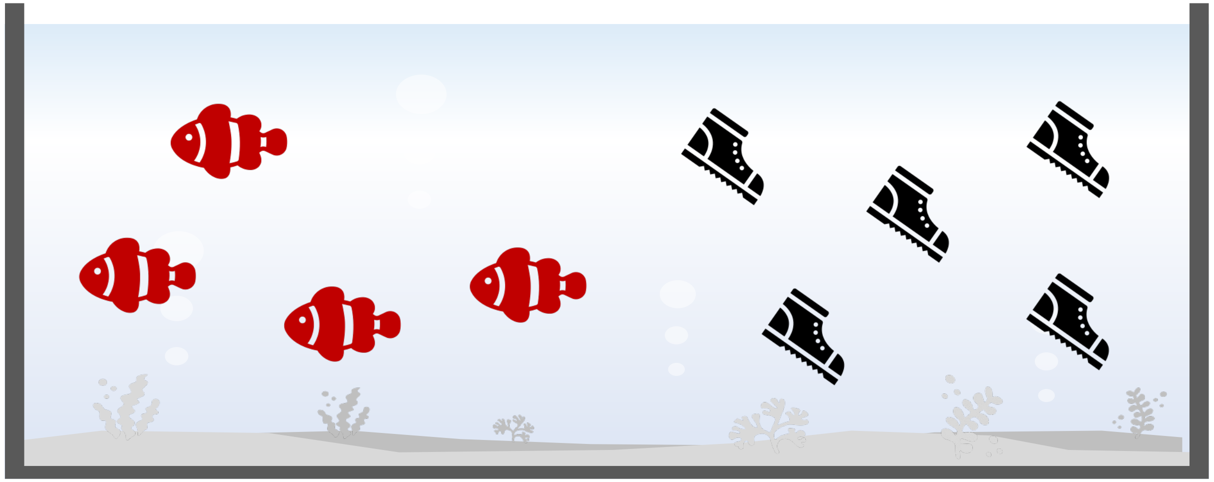






We need FISH





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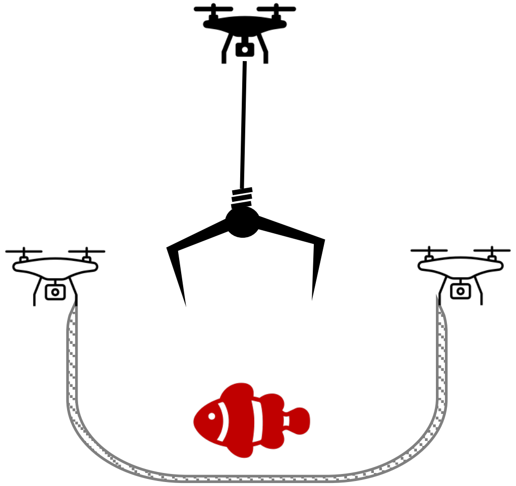
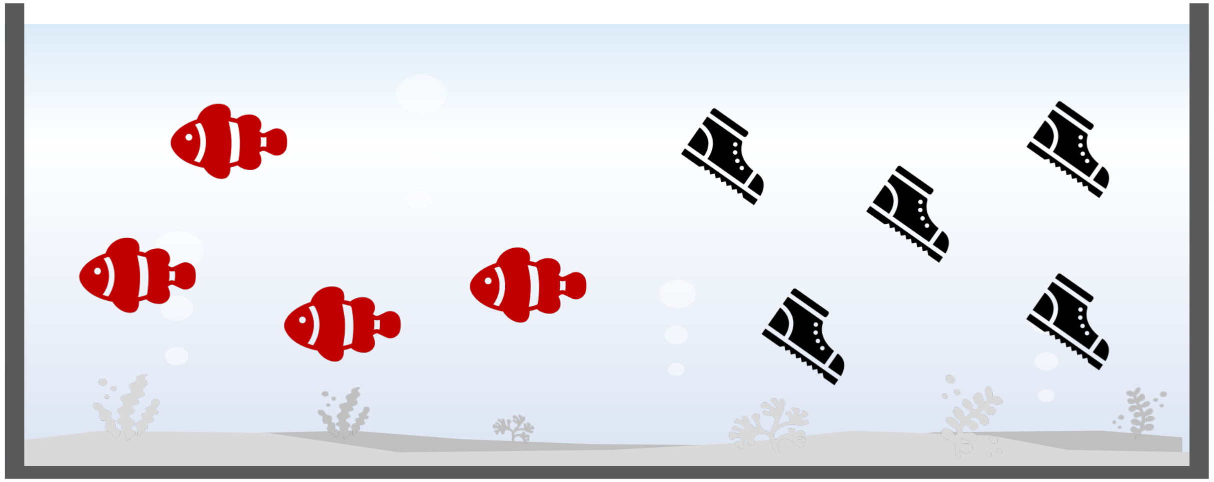
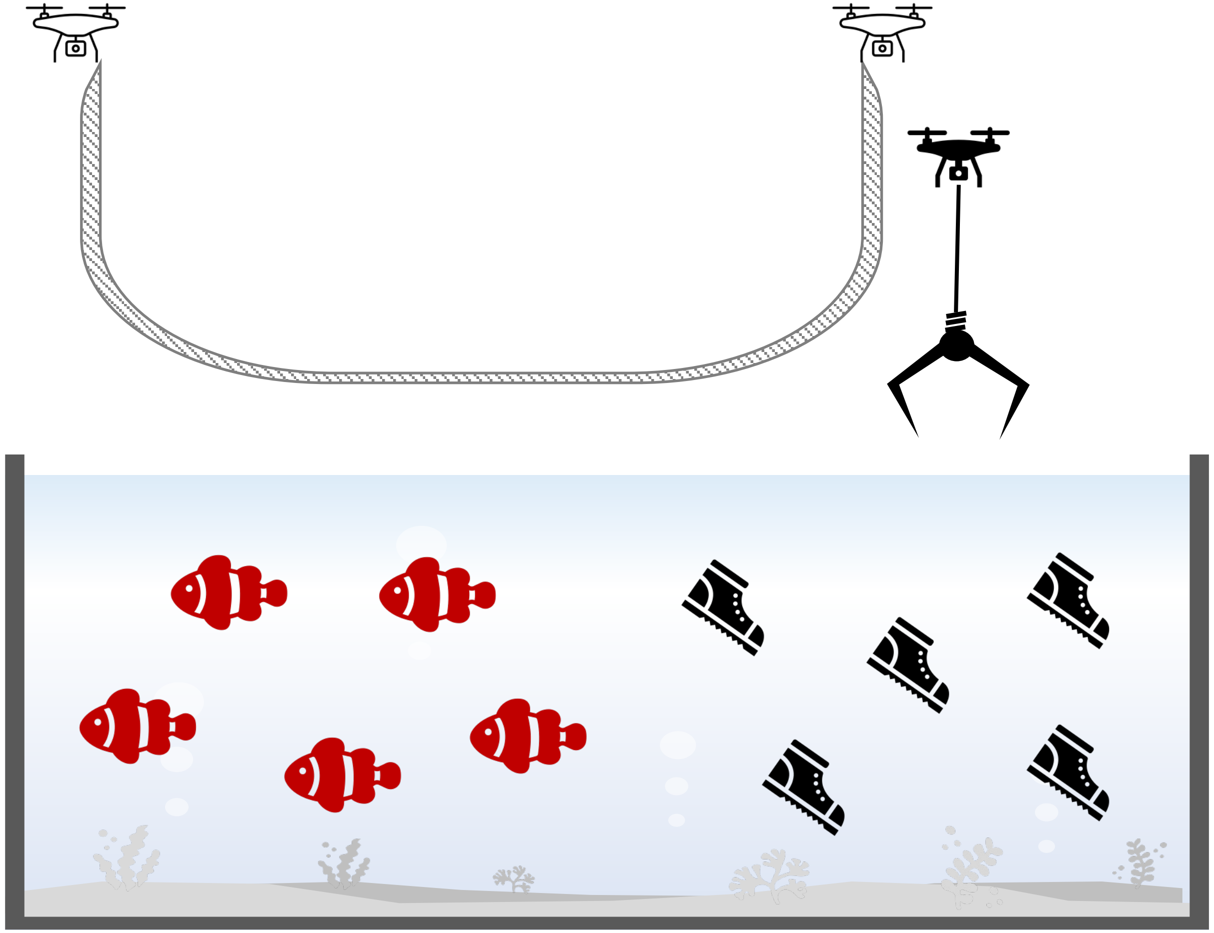


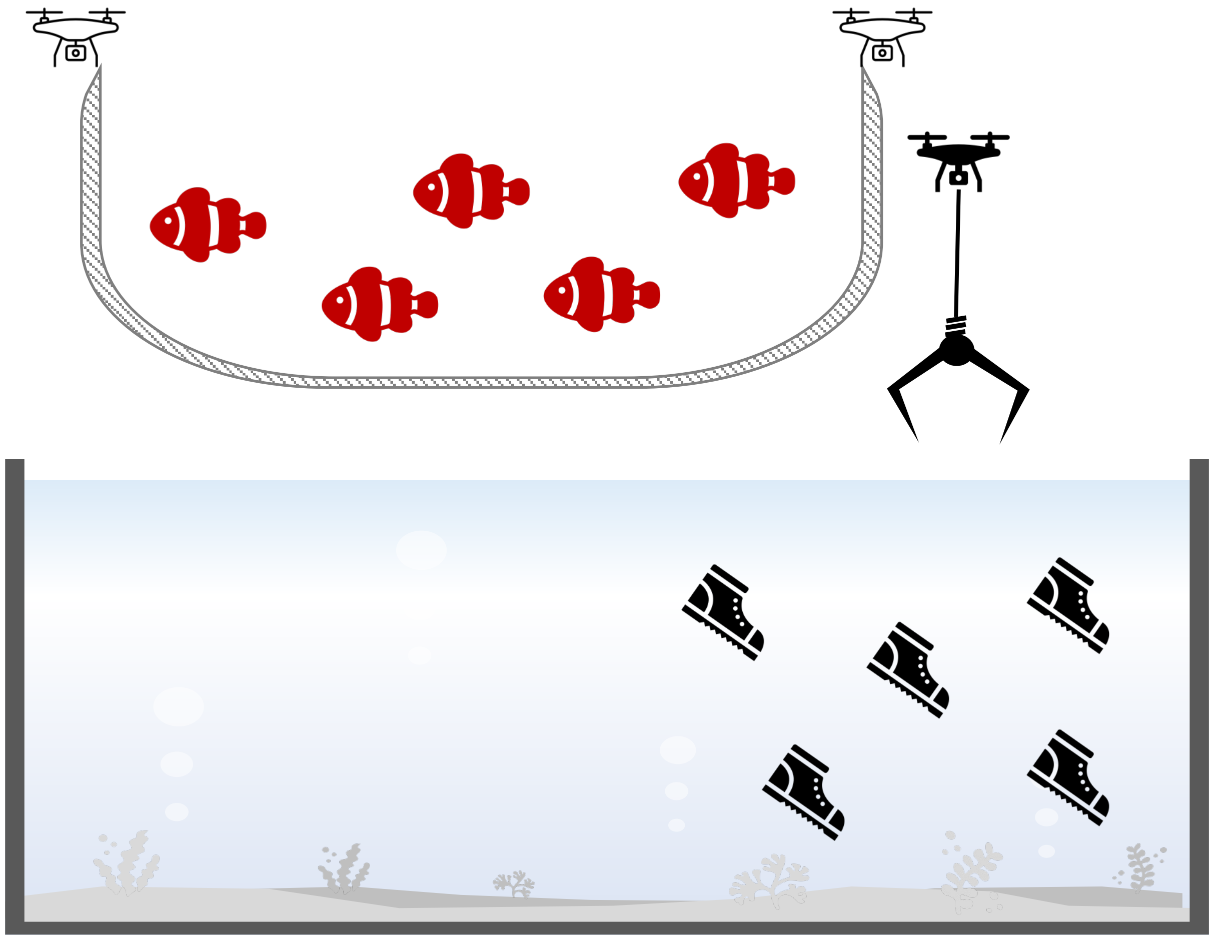


TP | TN | FP | FN



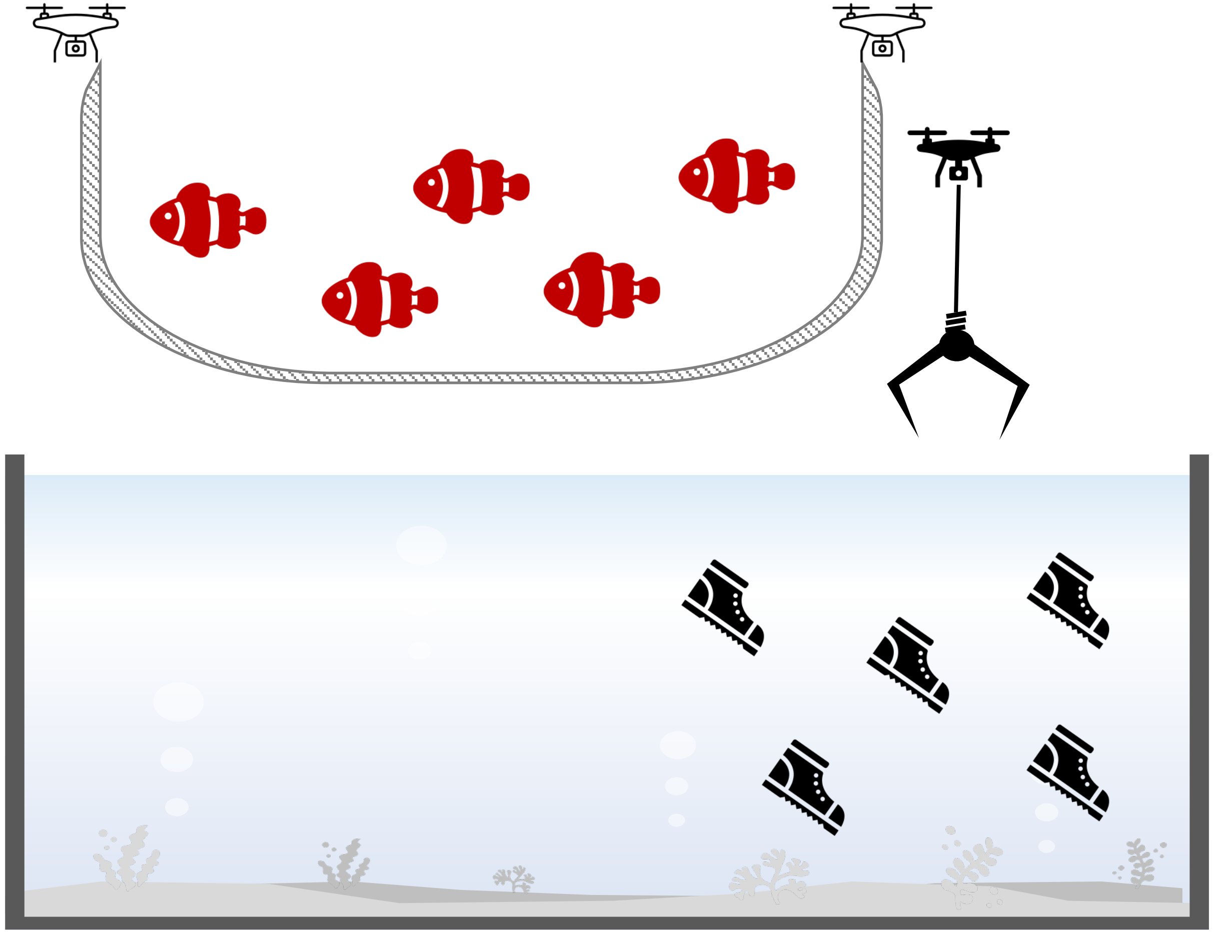
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

Ex. 1



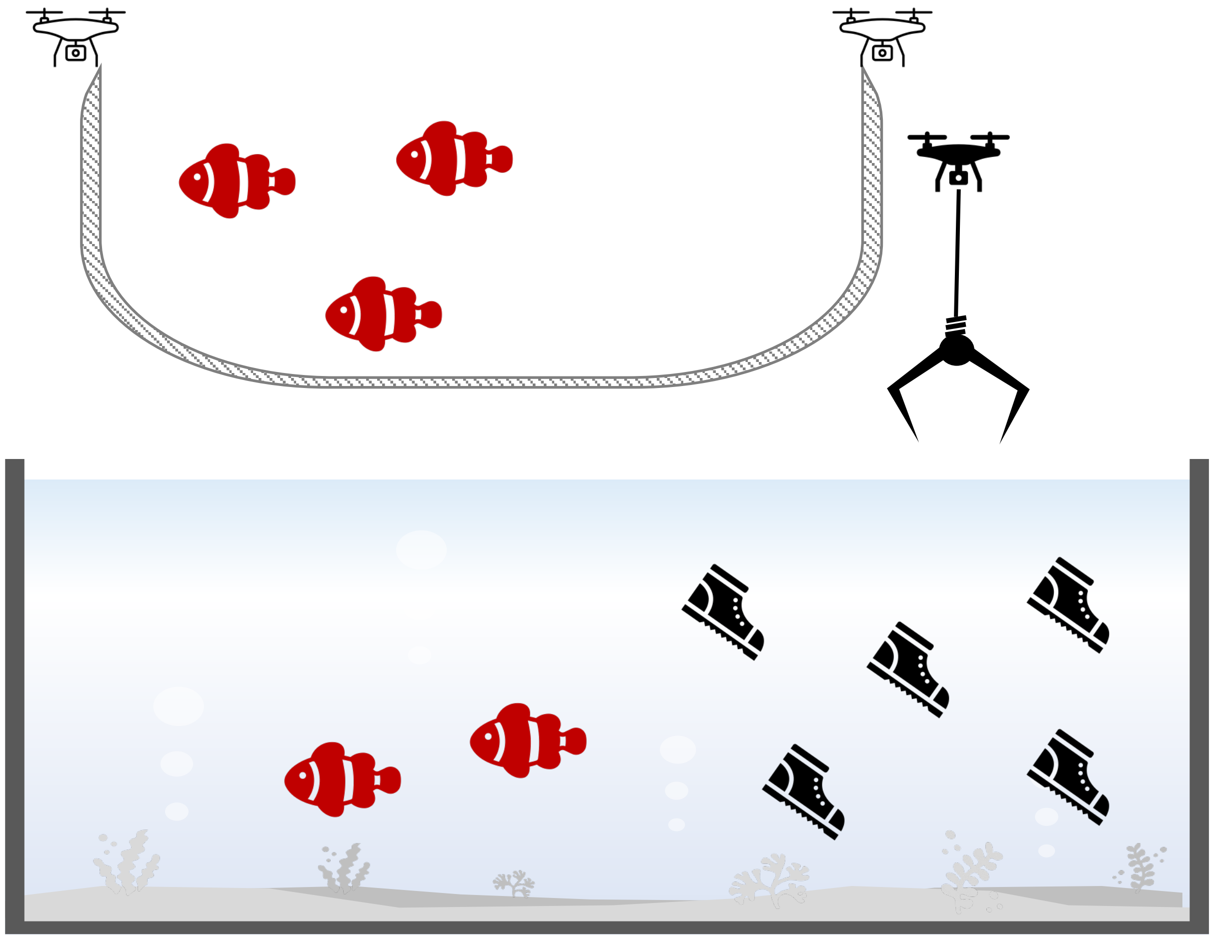
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

Ex. 1 (Ans)



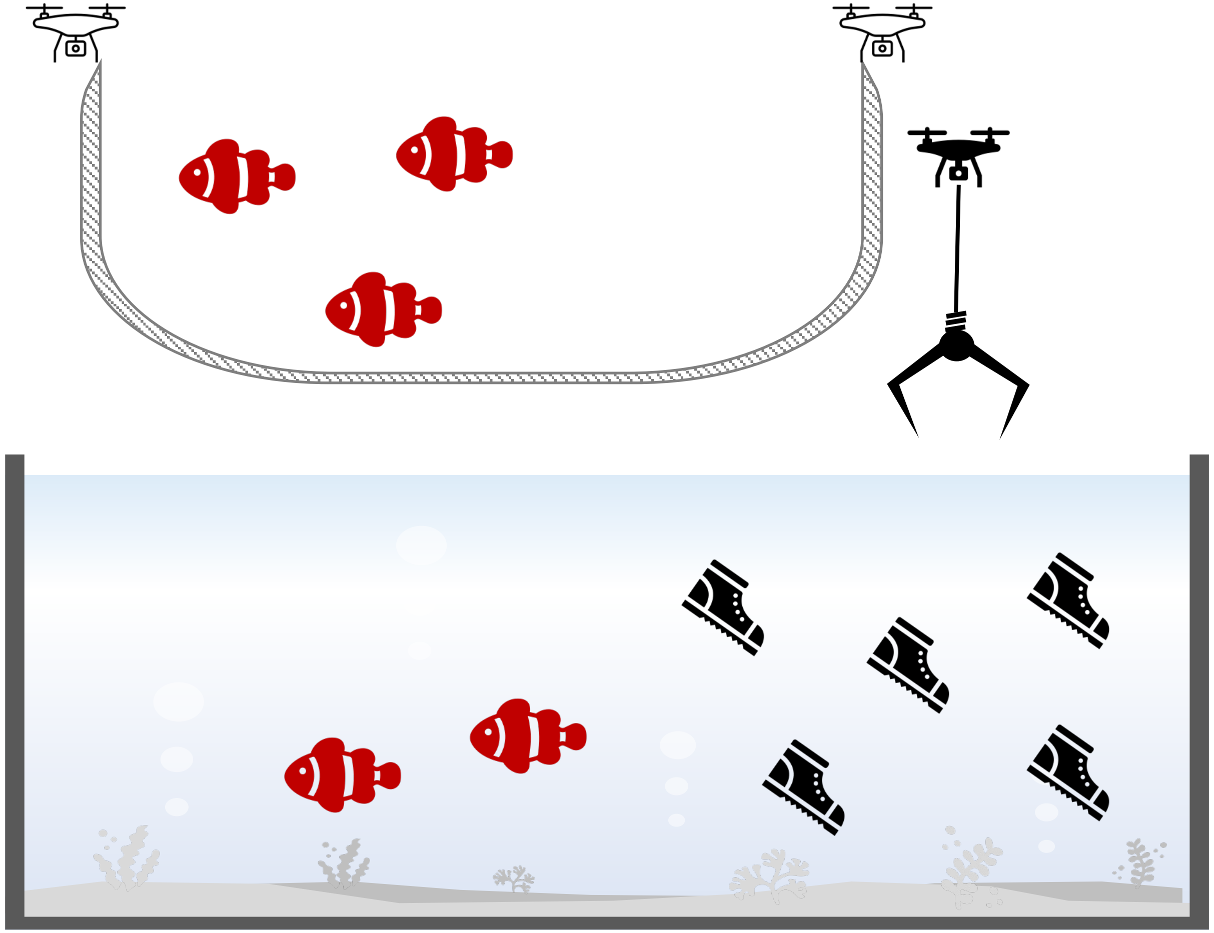
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

Ex. 2



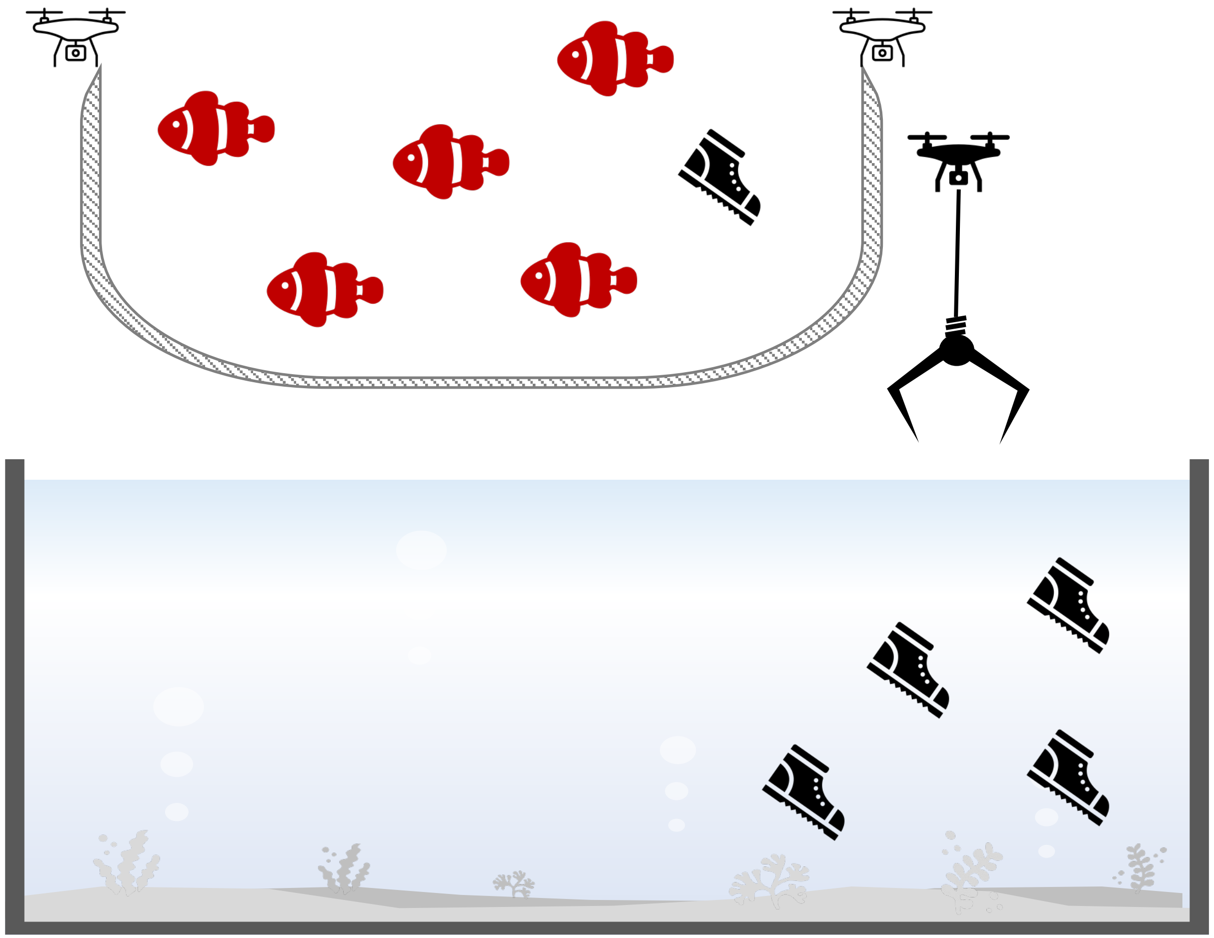
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

Ex. 2 (Ans)



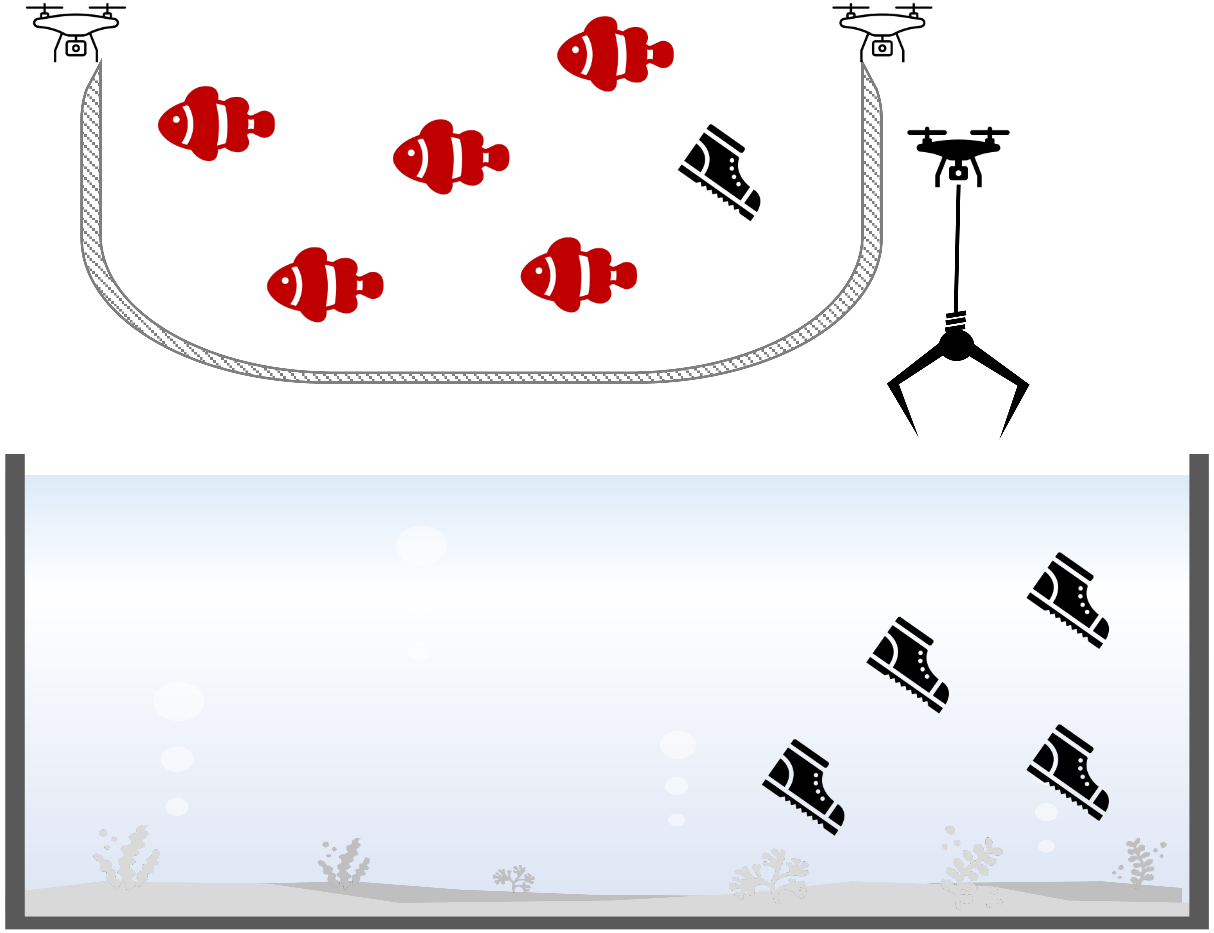
	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

Ex. 3



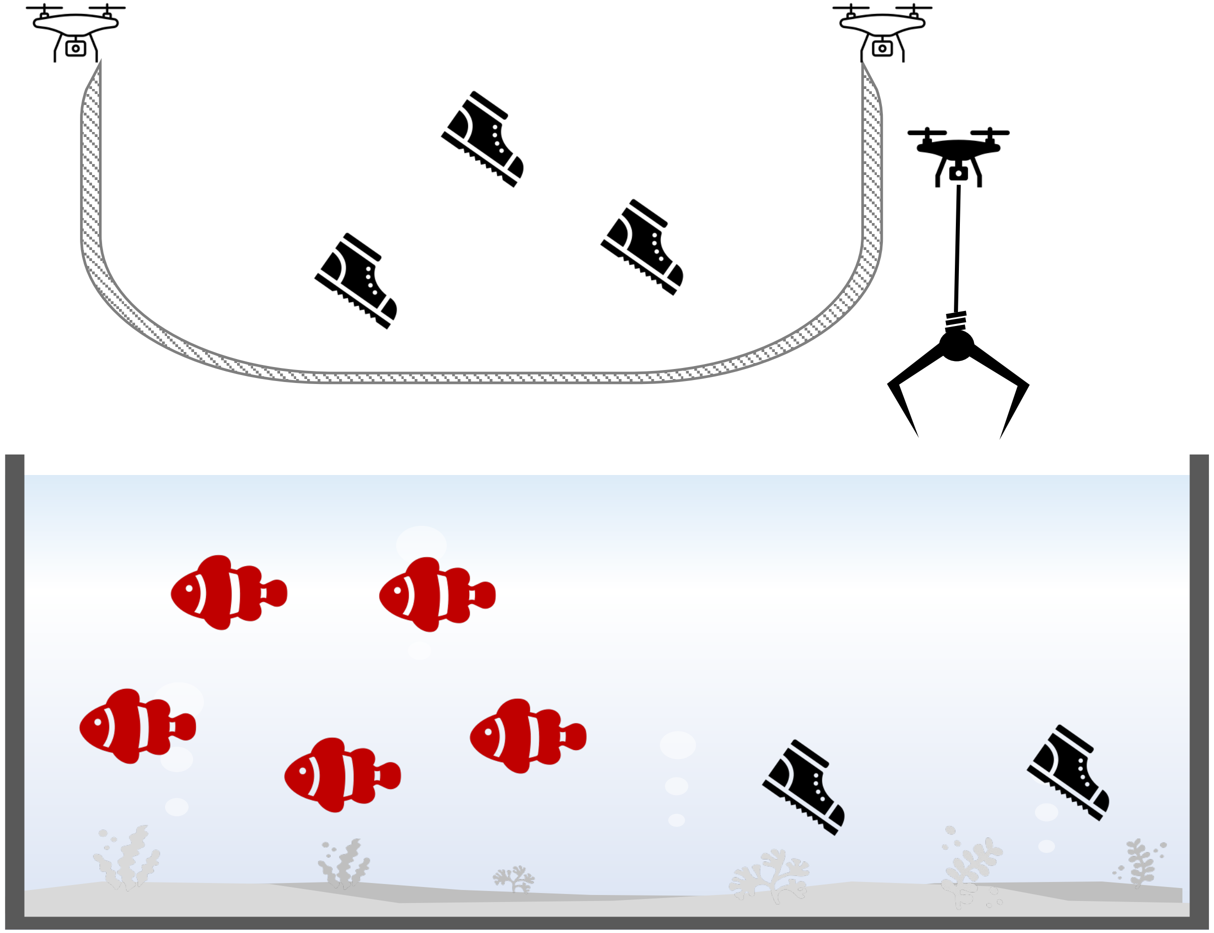
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

Ex. 3 (Ans)



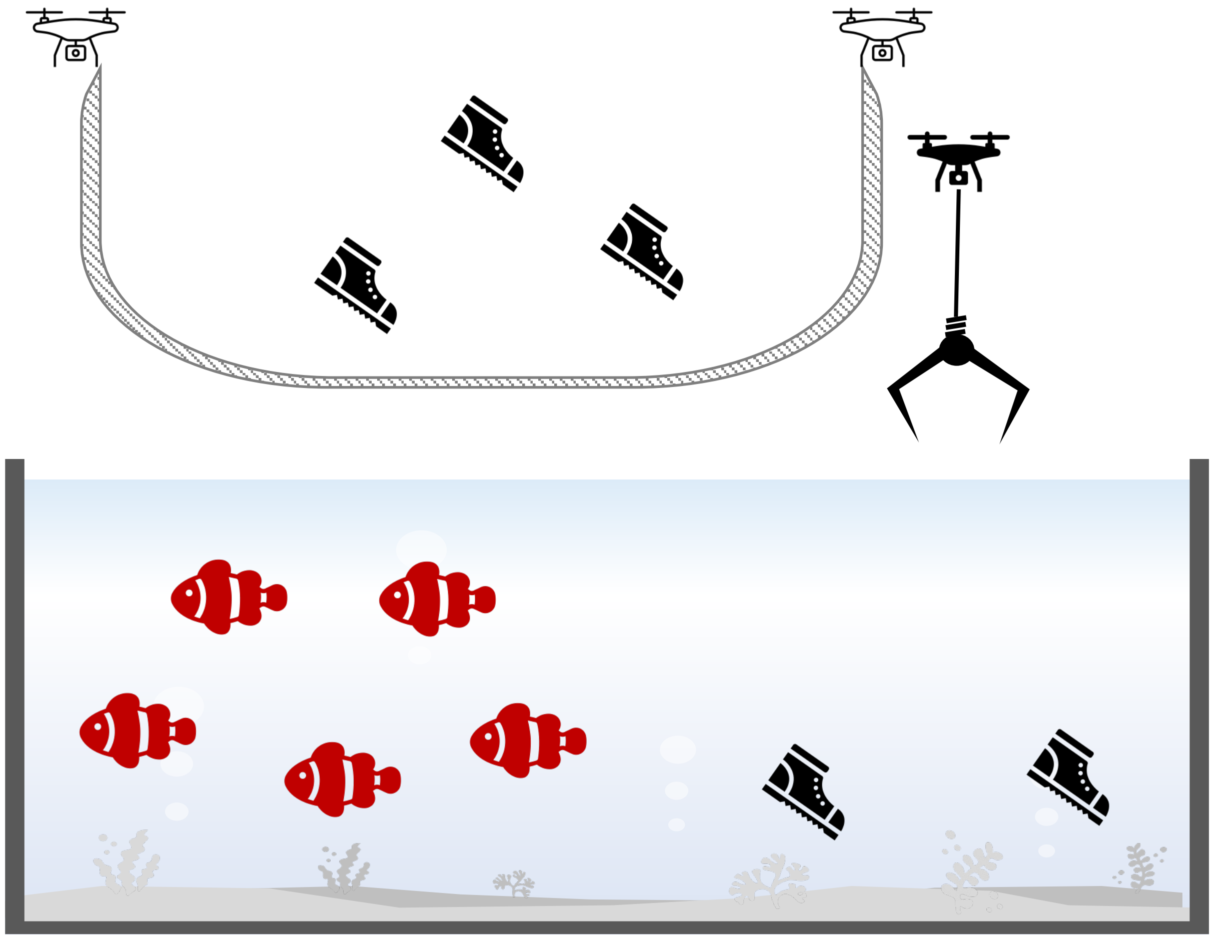
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**4** *(true-negative)*

Ex. 4



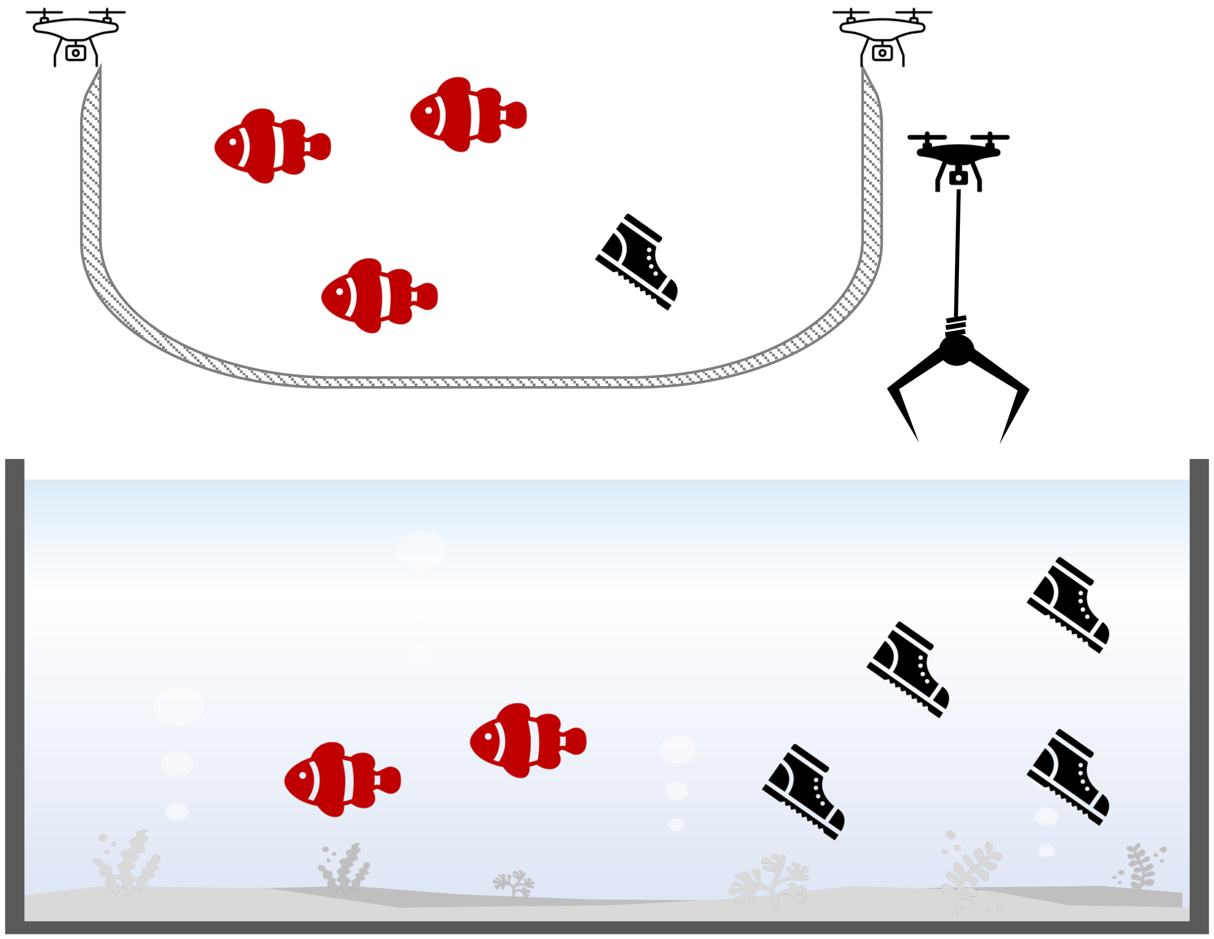
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

Ex. 4 (Ans)



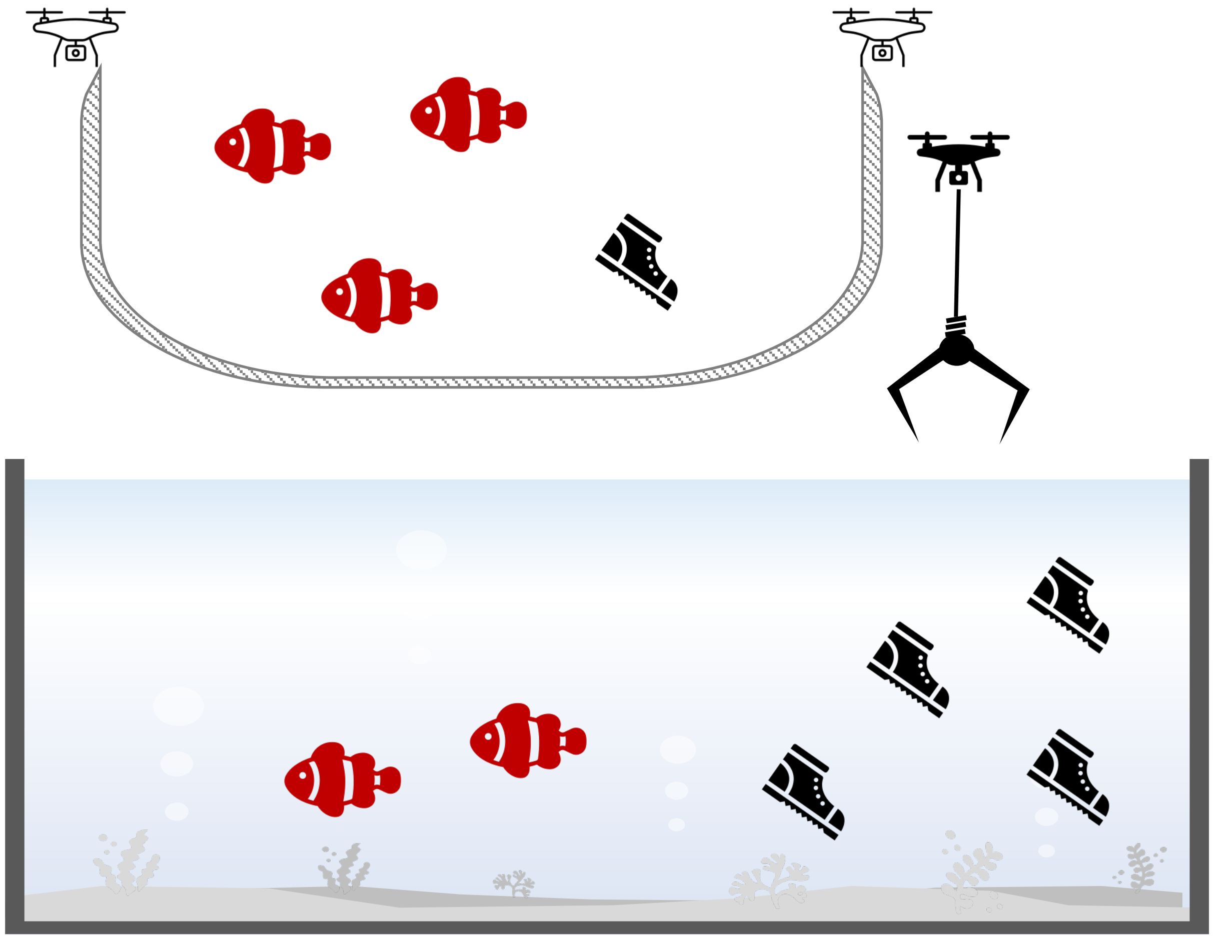
	predicted: **YES**	predicted: **no**
actual: **YES**	**0** *(true-positive)*	**5** *(false-negative)*
actual: **no**	**3** *(false-positive)*	**2** *(true-negative)*

Ex. 5



	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

Ex. 5 (Ans)



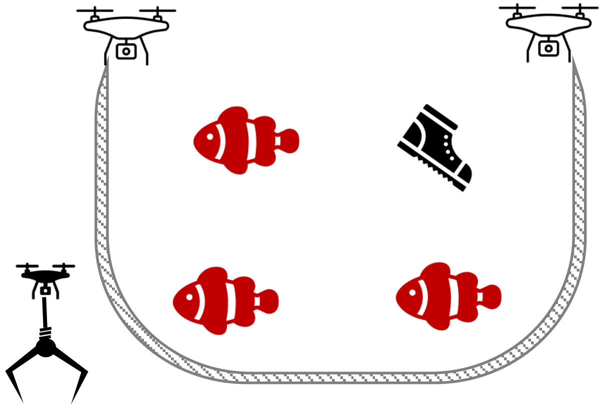
	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**4** *(true-negative)*

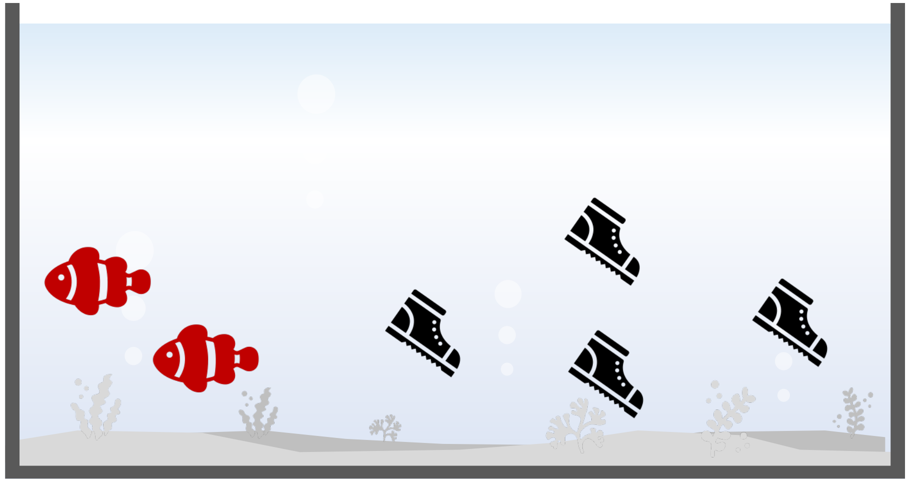
Accuracy

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 1

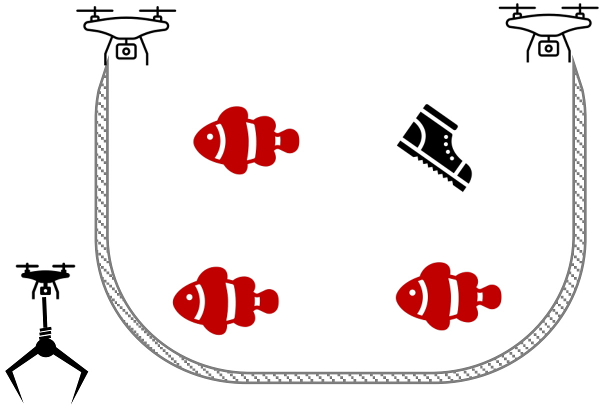




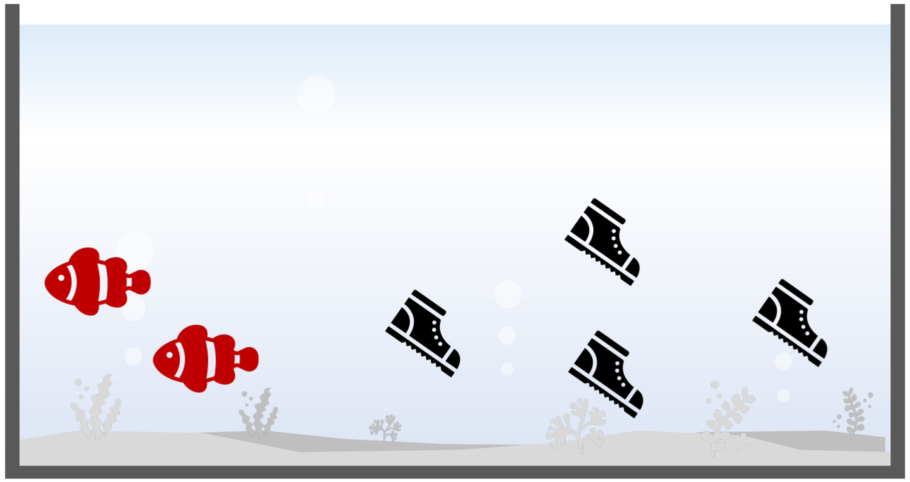
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 1

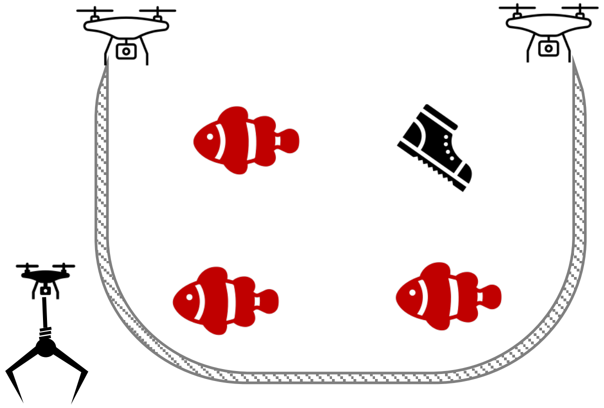


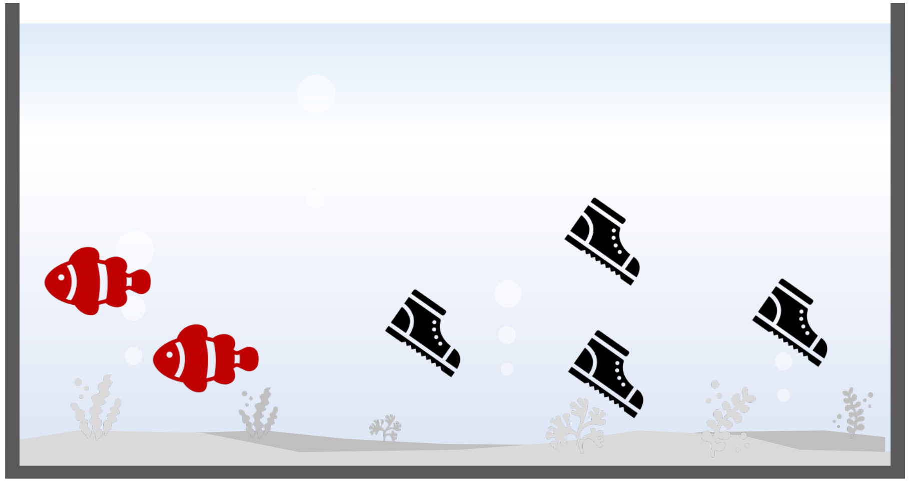
	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**4** *(true-negative)*



$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 1 (Ans)





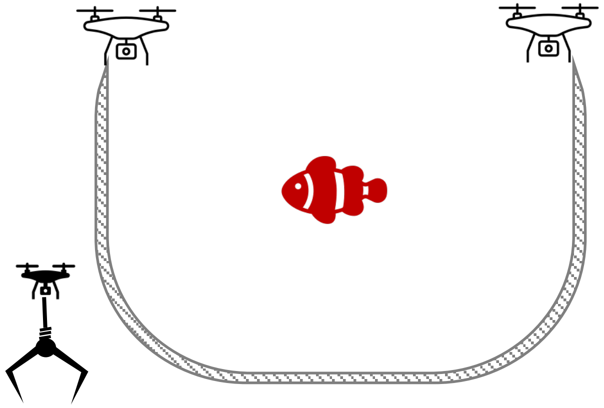
	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**4** *(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{3+4}{10}$$

$$= 0.7$$

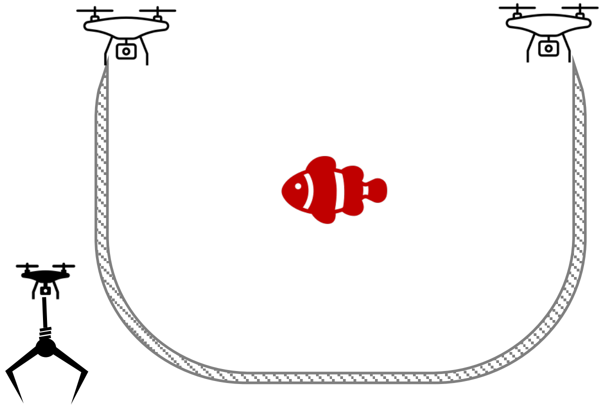
Ex. 2



	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 2 (Ans)



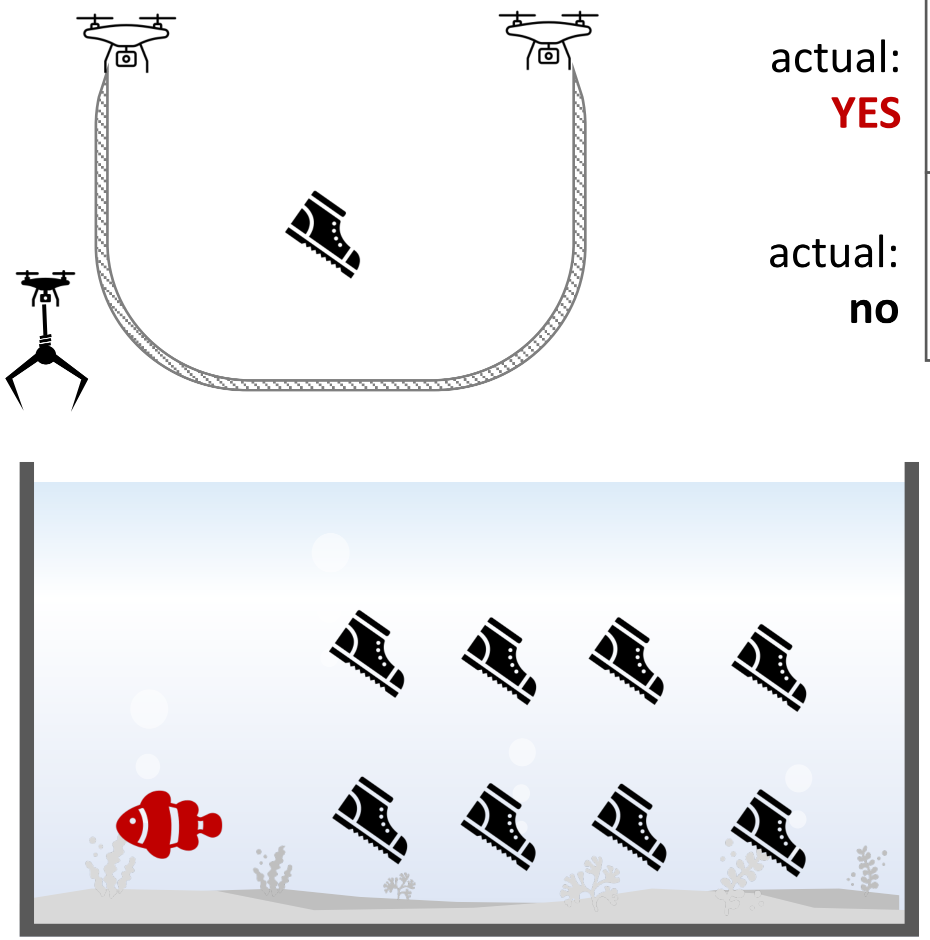
	predicted: **YES**	predicted: **no**
actual: **YES**	**1** *(true-positive)*	**1** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**8** *(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{1 + 8}{10}$$

$$= 0.9$$

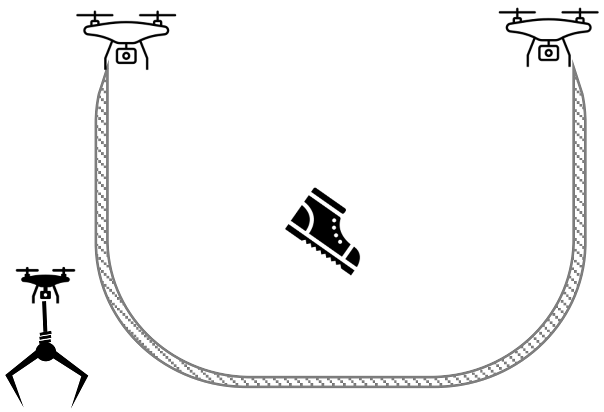
Ex. 3



	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 3 (Ans)



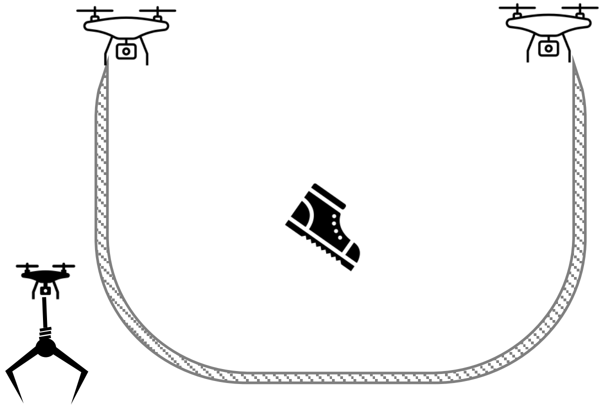
	predicted: **YES**	predicted: **no**
actual: **YES**	**0** *(true-positive)*	**1** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**8** *(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{0 + 8}{10}$$

$$= 0.8$$

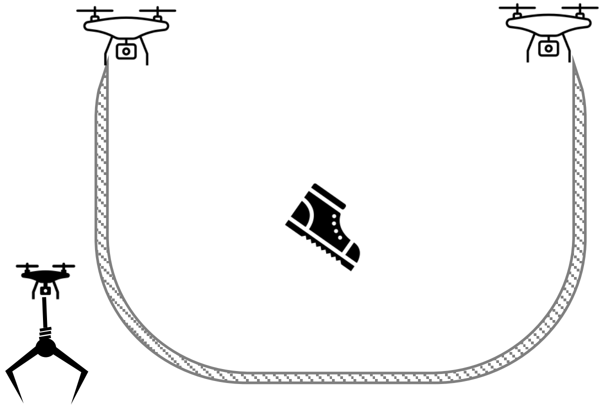
Ex. 4



	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 4 (Ans)



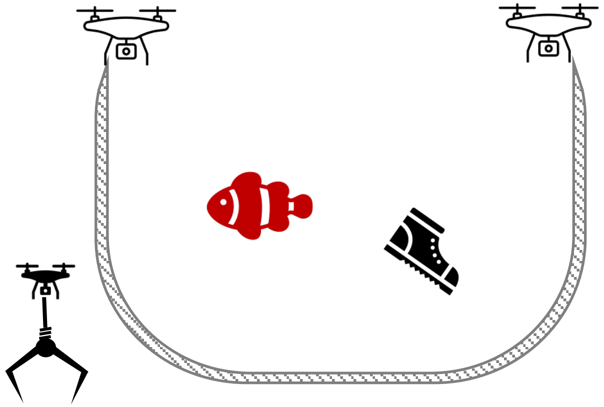
	predicted: **YES**	predicted: **no**
actual: **YES**	**0** *(true-positive)*	**3** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**96** *(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{0 + 96}{100}$$

$$= 0.96$$

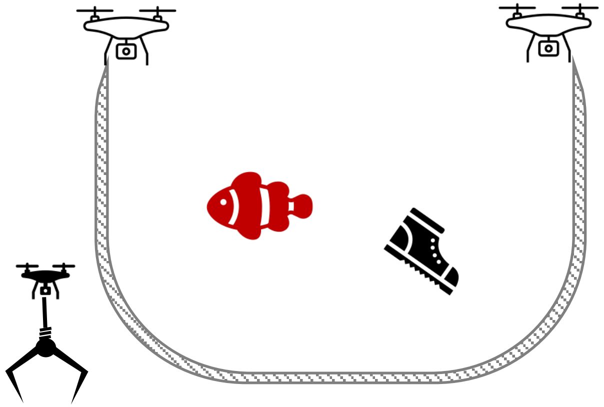
Ex. 5



	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex. 5 (Ans)



	predicted: **YES**	predicted: **no**
actual: **YES**	**1** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**96** *(true-negative)*

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{1 + 96}{100}$$

$$= 0.97$$

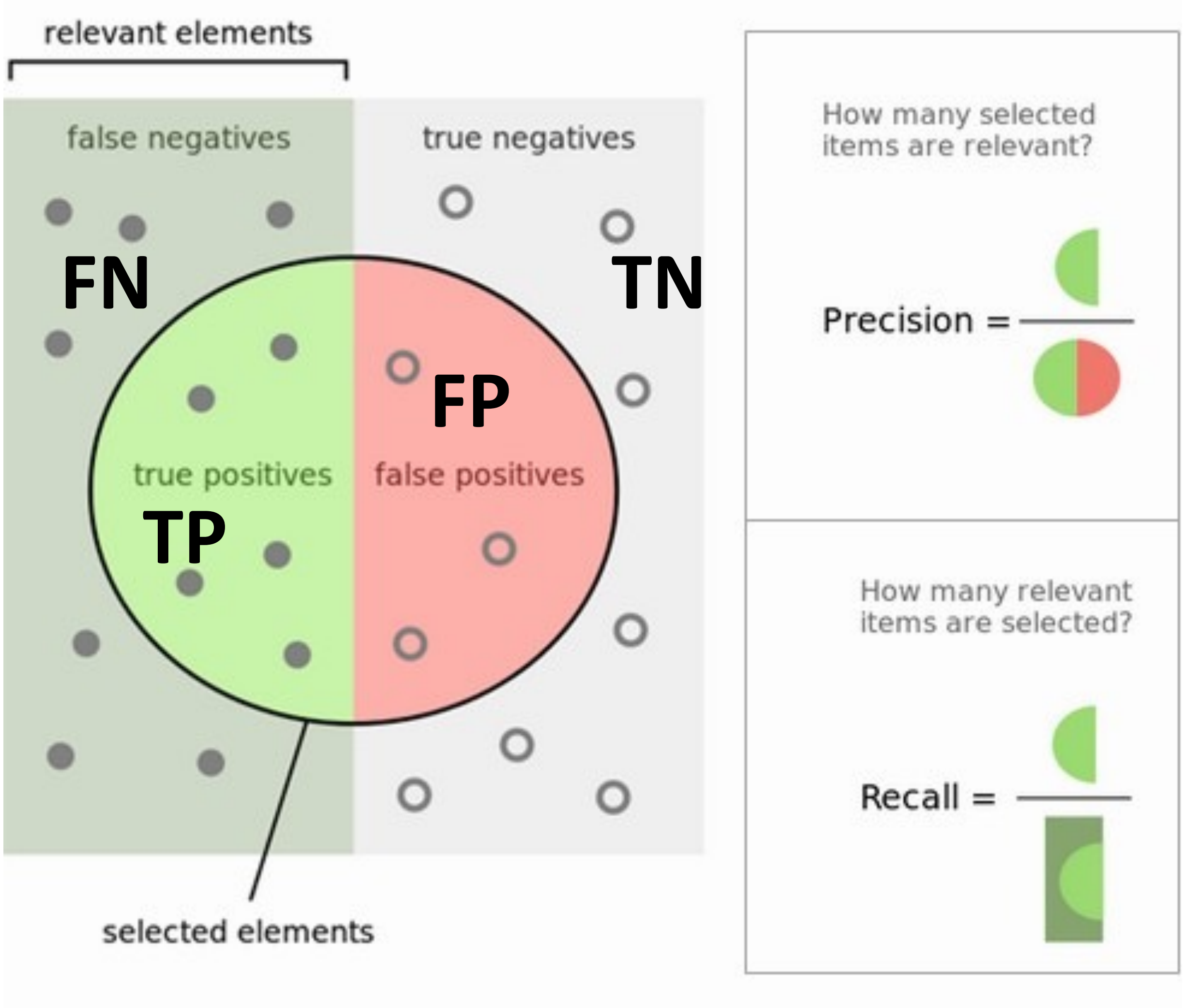
Precision | Recall | F1

Precision & Recall

$$Precision = \frac{TP}{TP + FP}$$

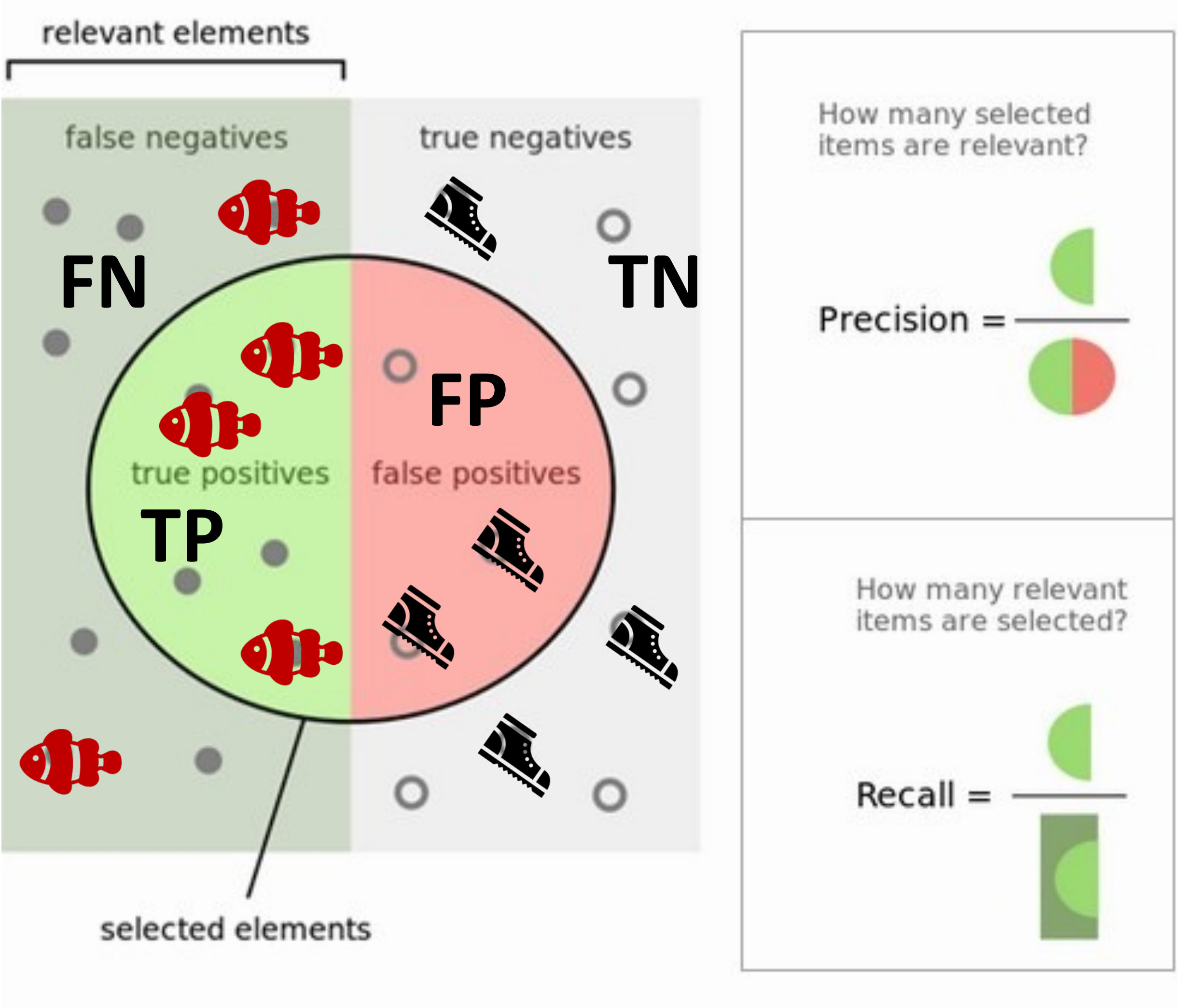
$$Recall = \frac{TP}{TP + FN}$$

Precision & Recall



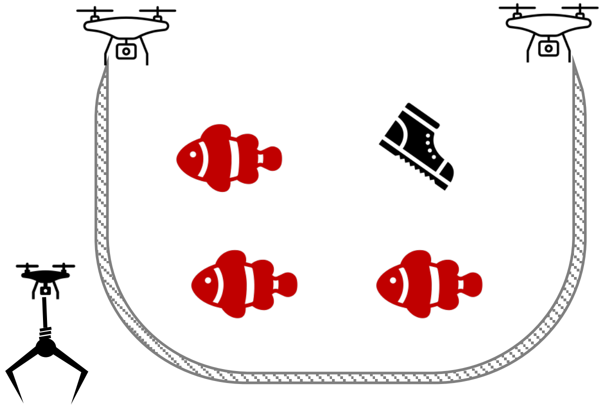
Ref: https://en.wikipedia.org/wiki/Precision_and_recall

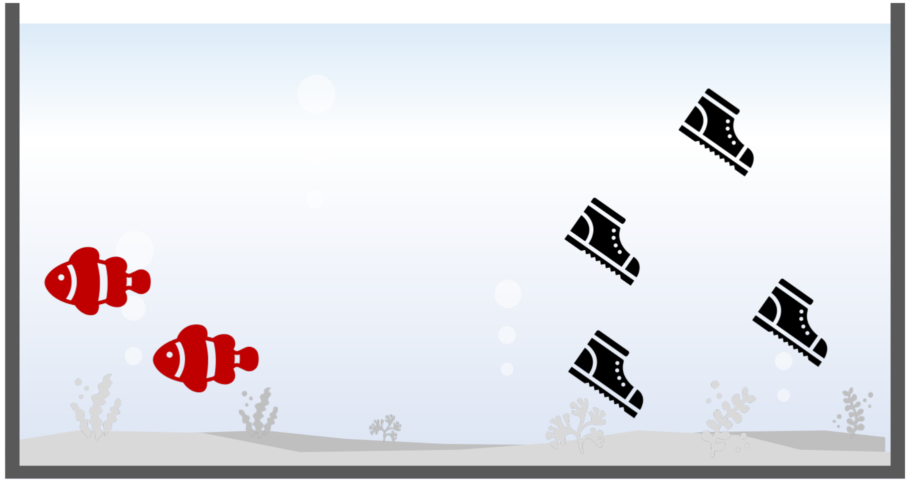
Precision & Recall



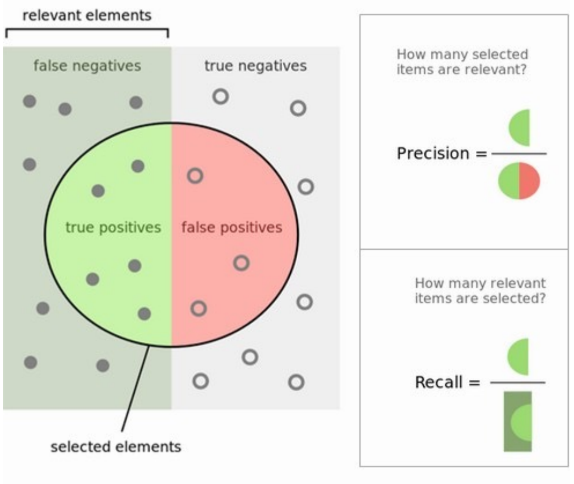
Ref: https://en.wikipedia.org/wiki/Precision_and_recall

Precision & Recall





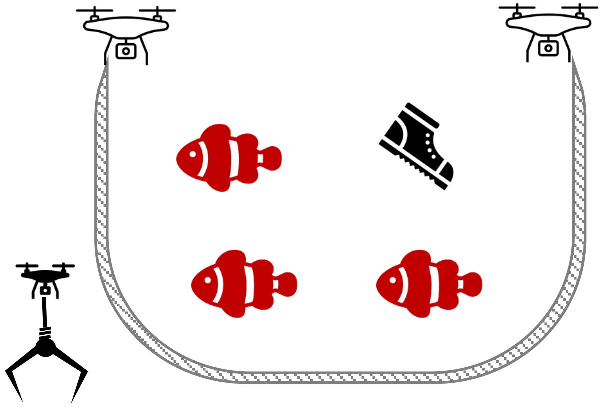
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*



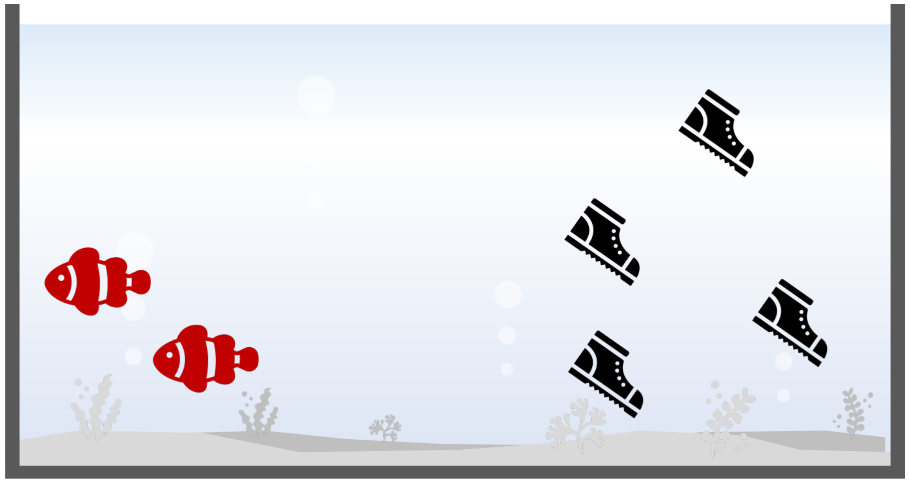
$$Precision = \frac{TP}{TP + FP}$$

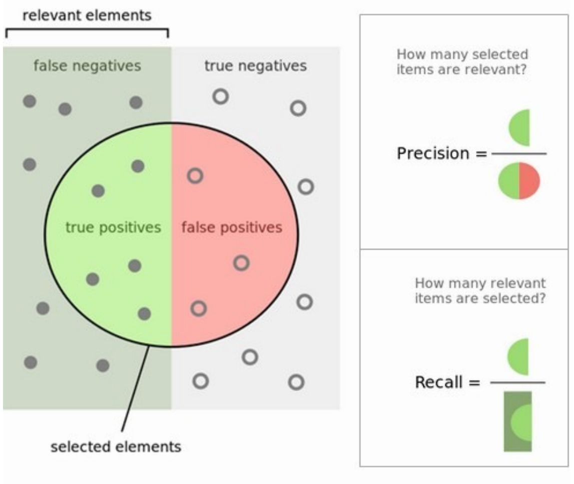
$$Recall = \frac{TP}{TP + FN}$$

Precision & Recall (Ans)



	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**4** *(true-negative)*





$$Precision = \frac{TP}{TP + FP} = \frac{3}{3+1}$$

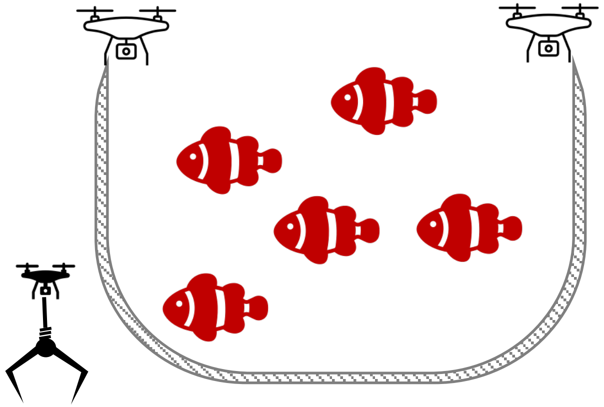
$$Recall = \frac{TP}{TP + FN} = \frac{3}{3+2}$$

Questions

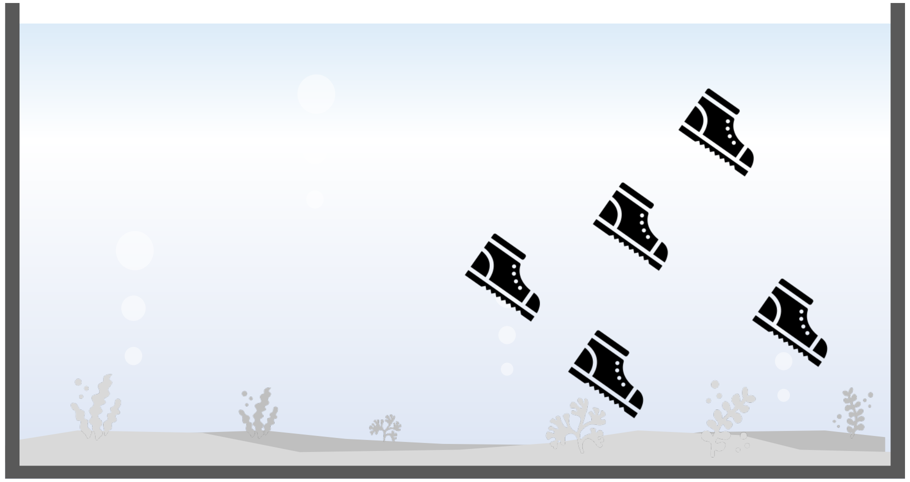
Precision ≈ 1.00

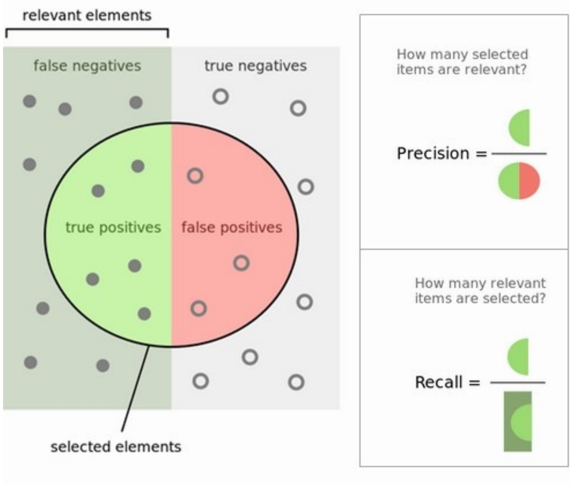
is it good?

Ex. 1.1



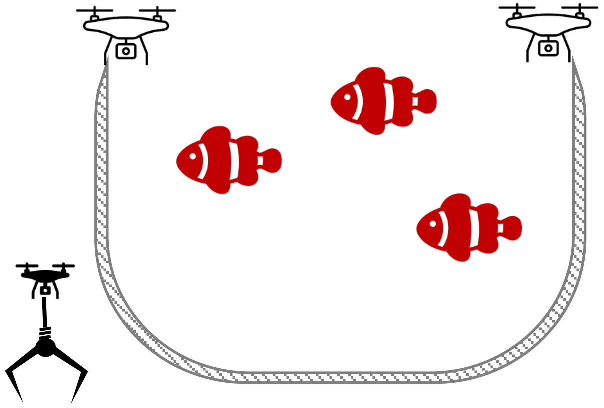
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*



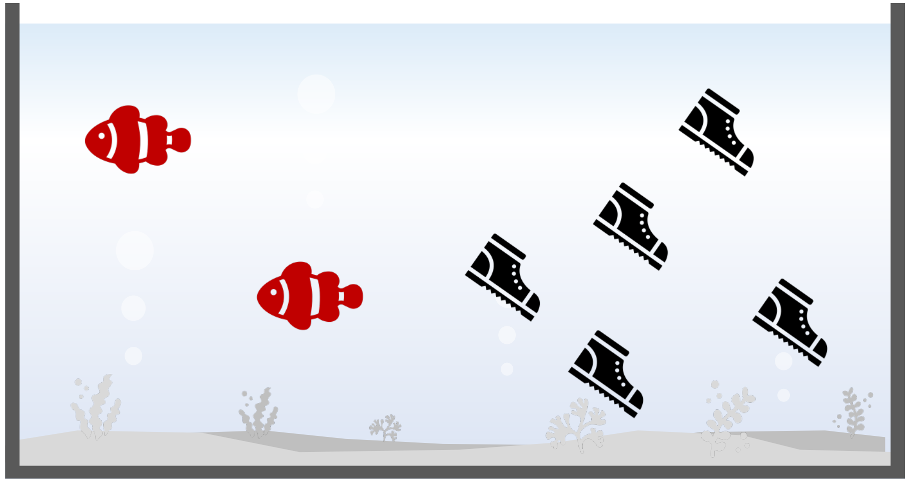


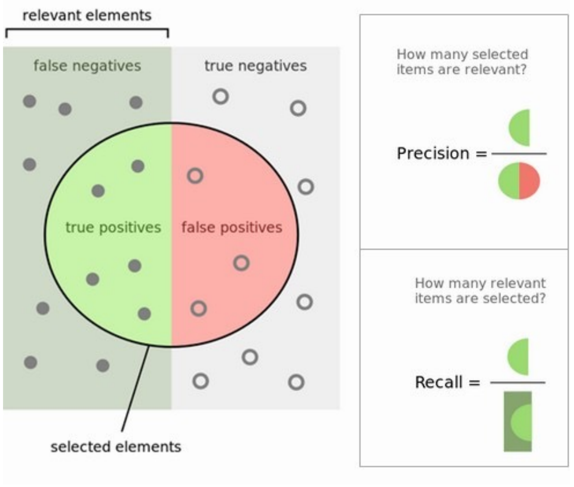
$$Precision = \frac{TP}{TP + FP} = \frac{5}{5 + 0}$$

Ex. 1.2



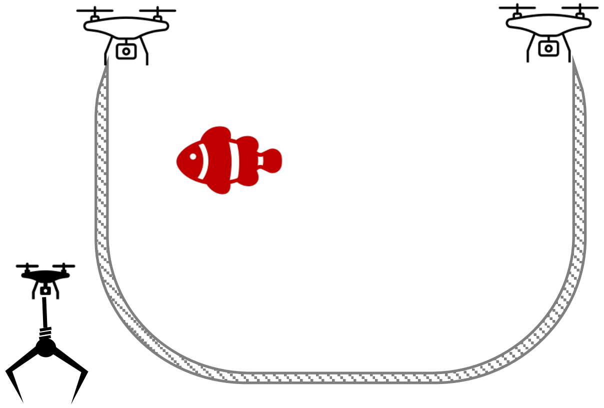
	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*



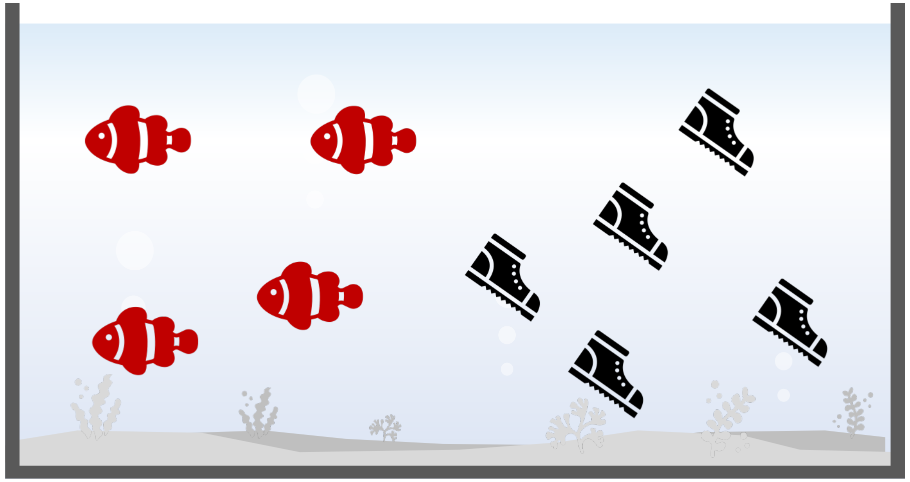


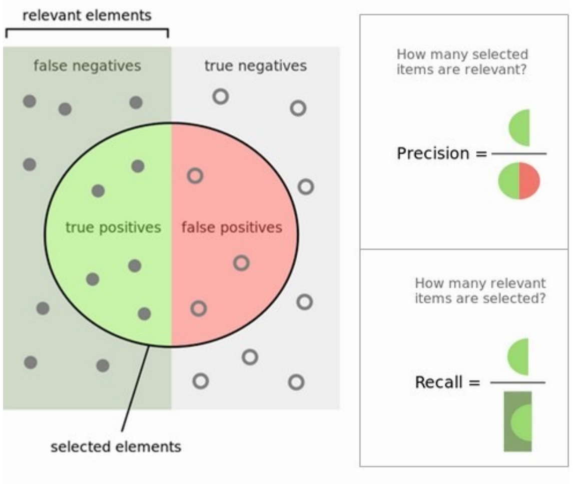
$$Precision = \frac{TP}{TP + FP} = \frac{3}{3 + 0}$$

Ex. 1.2



	predicted: **YES**	predicted: **no**
actual: **YES**	**1** *(true-positive)*	**4** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*





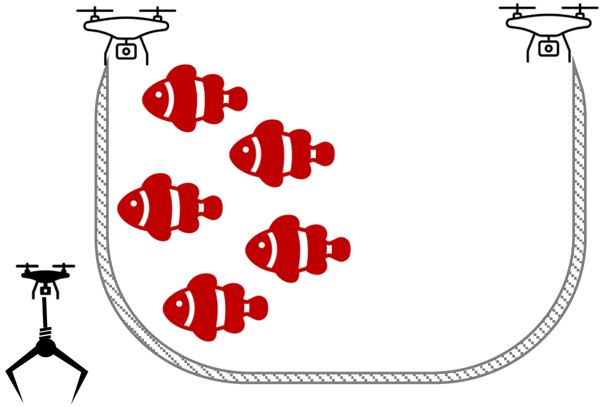
$$Precision = \frac{TP}{TP + FP} = \frac{1}{1 + 0}$$

Questions

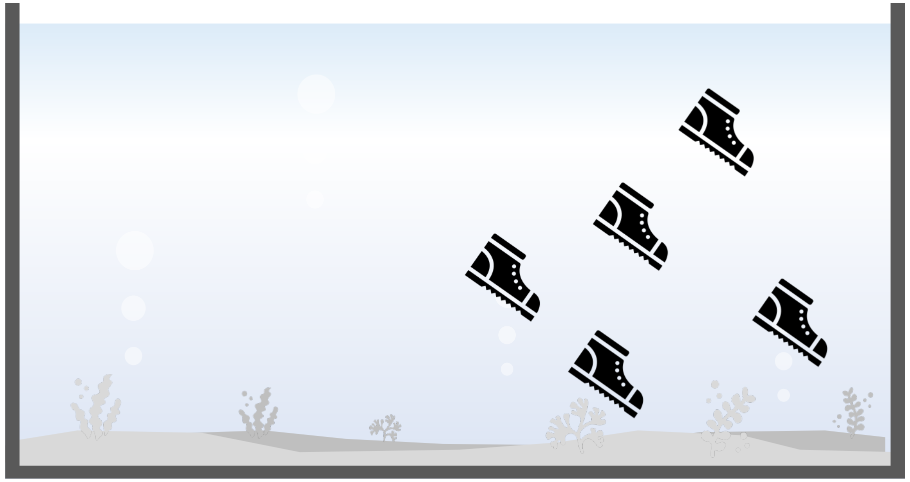
Recall ≈ 1.00

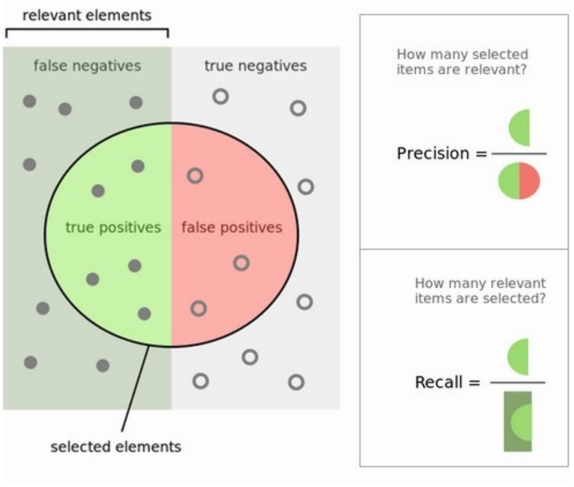
is it good?

Ex. 2.1



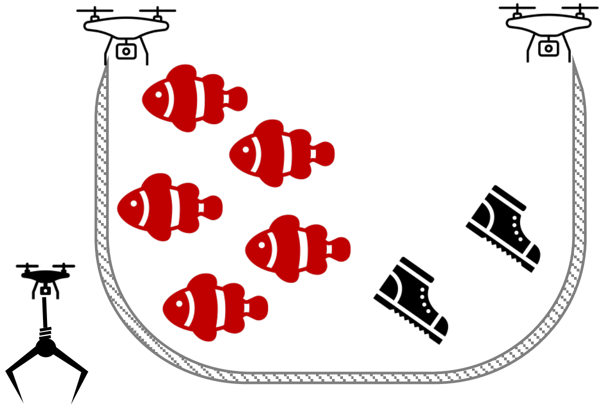
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*



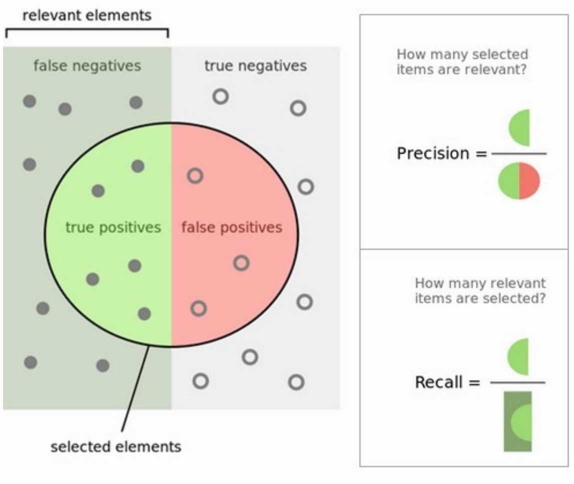


$$Recall = \frac{TP}{TP + FN} = \frac{5}{5 + 0}$$

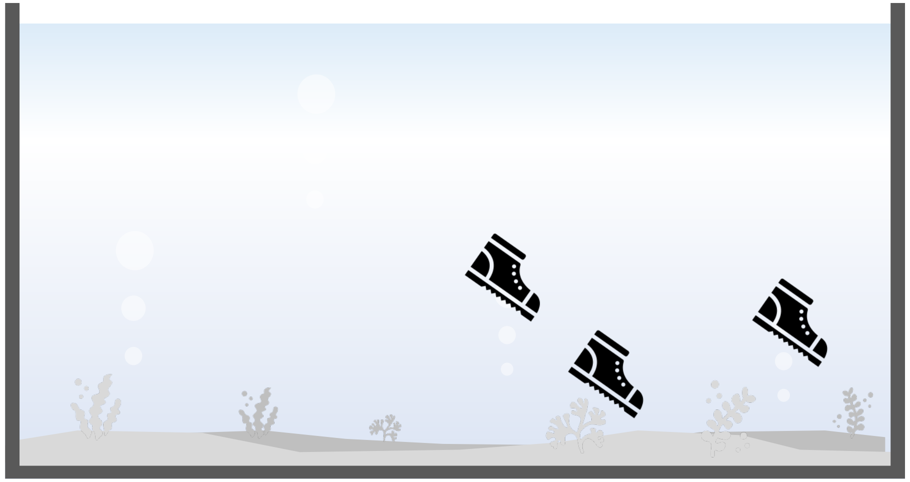
Ex. 2.2



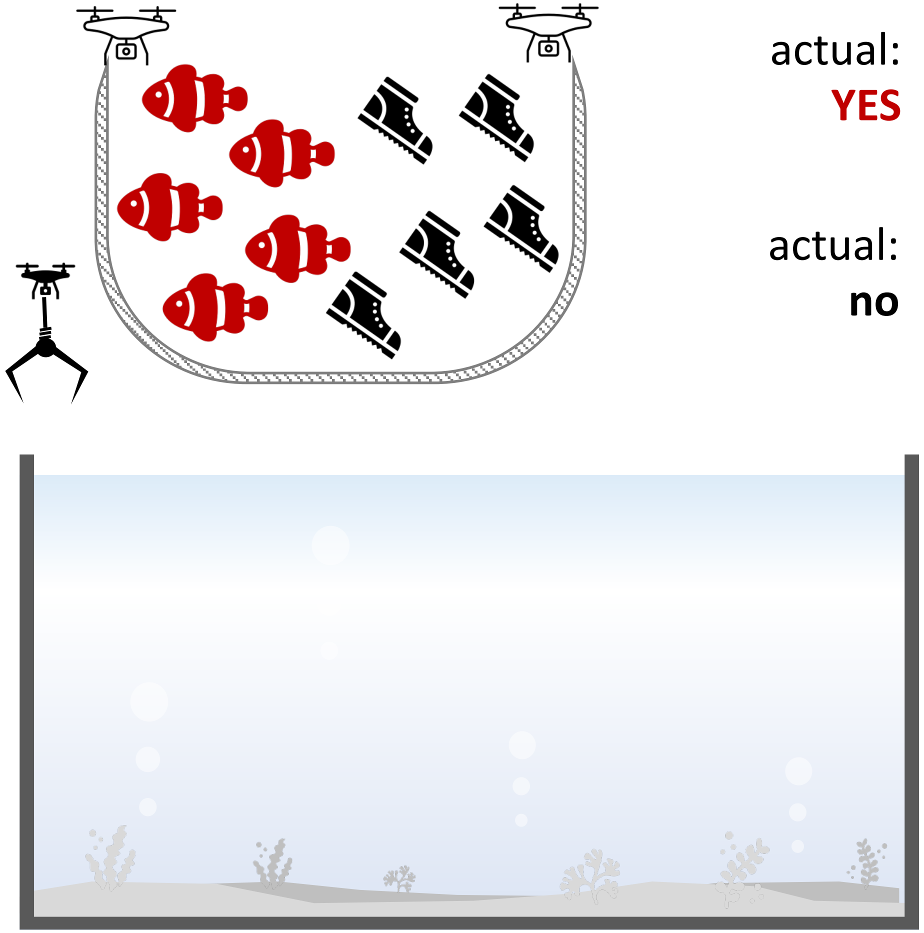
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**2** *(false-positive)*	**3** *(true-negative)*



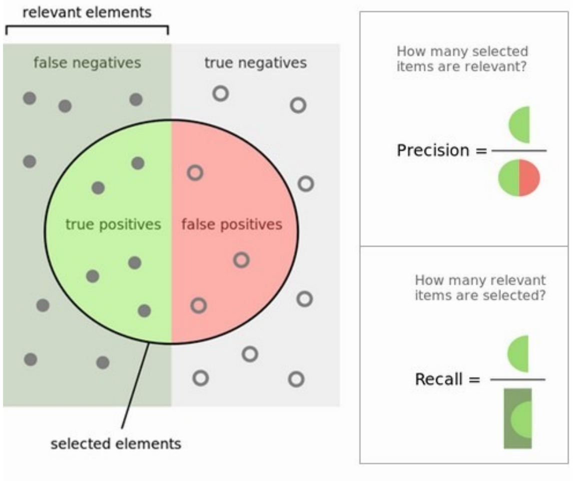
$$Recall = \frac{TP}{TP + FN} = \frac{5}{5+0}$$



Ex. 2.3



	predicted: YES	predicted: no
actual: YES	5 (true-positive)	0 (false-negative)
actual: no	5 (false-positive)	0 (true-negative)



$$Recall = \frac{TP}{TP + FN} = \frac{5}{5 + 0}$$

F1 (F-Measure)

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2\times \frac{Precision \times Recall}{Precision + Recall}$$

F1 (F-Measure)

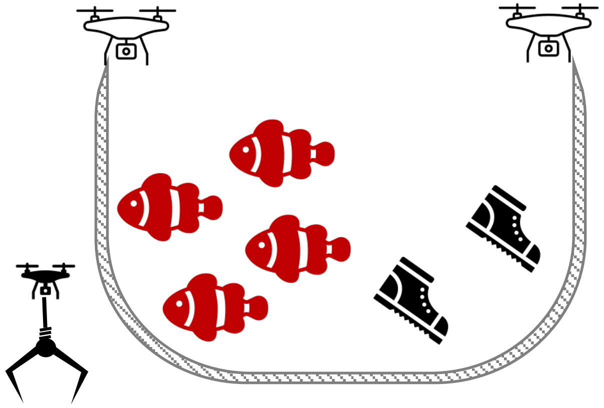
$$\frac{1}{F1} = \frac{\frac{1}{Precision} + \frac{1}{Recall}}{2}$$

$$F1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

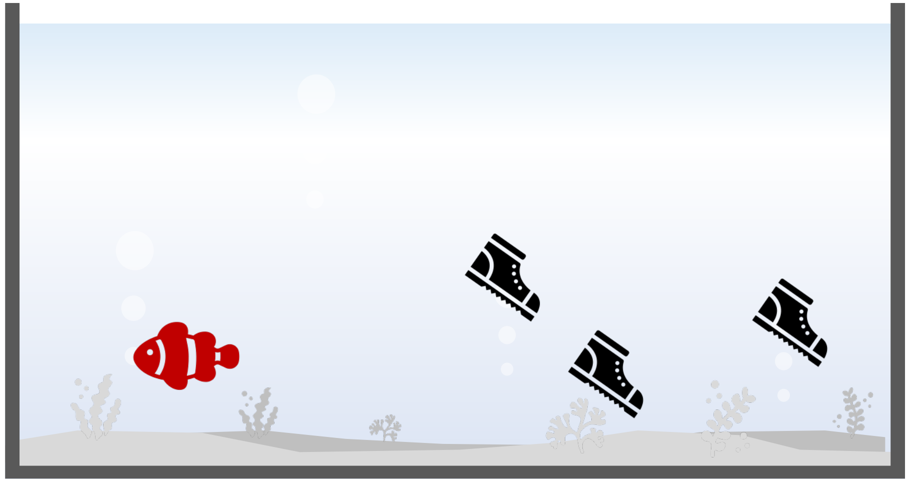
$$F1 = \frac{2}{\frac{Precision + Recall}{Precision \times Recall}}$$

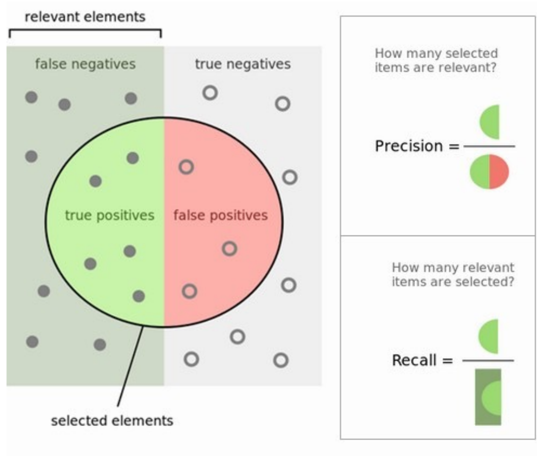
$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Ex. 3.1



	predicted: **YES**	predicted: **no**
actual: **YES**	**4** *(true-positive)*	**1** *(false-negative)*
actual: **no**	**2** *(false-positive)*	**3** *(true-negative)*





$$Precision = \frac{TP}{TP + FP} = \frac{4}{4+2} = 0.67$$

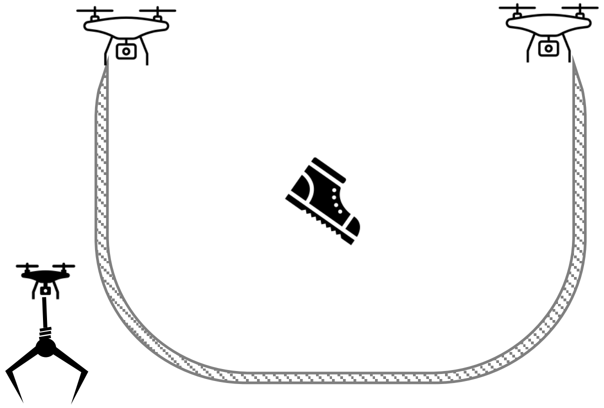
$$Recall = \frac{TP}{TP + FN} = \frac{4}{4+1} = 0.80$$

$$\boldsymbol{F1} = 2\times \frac{Precision \times Recall}{Precision + Recall}$$

$$= 2\times \frac{0.67 \times 0.80}{0.67 + 0.80}$$

$$= \mathbf{0.73}$$

Ex. 3.2



	predicted: **YES**	predicted: **no**
actual: **YES**	**0** *(true-positive)*	**3** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**96** *(true-negative)*

$$Accuracy = \frac{0+96}{100} = 0.96$$

$$Precision = \frac{TP}{TP+FP} = \frac{0}{1+1} = 0$$

$$Recall = \frac{TP}{TP+FN} = \frac{0}{1+2} = 0$$

$$\boldsymbol{F1} = 2\times \frac{Precision \times Recall}{Precision + Recall}$$

$$= 2\times \frac{0\times 0}{0+0}$$

$$= \mathbf{0}$$ (if zero_division = 0)

Recommendation

- Balanced Class Distribution
  - Accuracy
- Imbalanced Class Distribution
  - Precision, Recall, F1

Multi-Class ?

- E.g.
  - Class A → 10 samples
  - Class B → 10 samples
  - Class C → 10 samples
  - Class D → 10 samples
- Balanced ?

Multi-Class ?

- E.g.
  - Class A → 10 samples
  - Class B → 10 samples
  - Class C → 10 samples
  - Class D → 10 samples
- Balanced ?

- Think Carefully about Binary Classifcation
  - Class A (10) vs. Not-Class A (30)
  - Class B (10) vs. Not-Class B (30)
  - Class C (10) vs. Not-Class C (30)
  - Class D (10) vs. Not-Class D (30)
- Balanced?

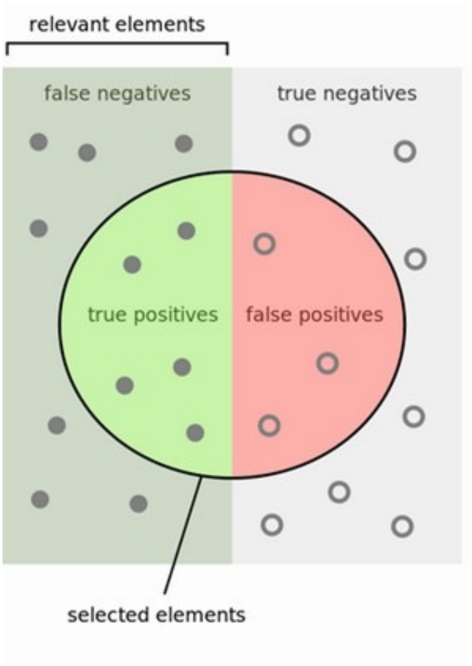
Multi-Class ?

- E.g.
  - Class A → 10 samples
  - Class B → 10 samples
  - Class C → 10 samples
  - Class D → 10 samples
- Balanced ?

- Think Carefully about Binary Classifcation
  - Class A (10) vs. Not-Class A (30)
  - Class B (10) vs. Not-Class B (30)
  - Class C (10) vs. Not-Class C (30)
  - Class D (10) vs. Not-Class D (30)
- Balanced? → Imbalanced
- Recommendation
  - Precision, Recall, F1

TPR | FPR | ROC | AUC

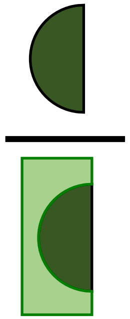
TP Rate & FP Rate



	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

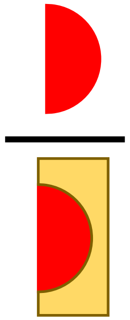
- True-Positive Rate

$$\boldsymbol{TPR} = \frac{TP}{TP + FN}$$

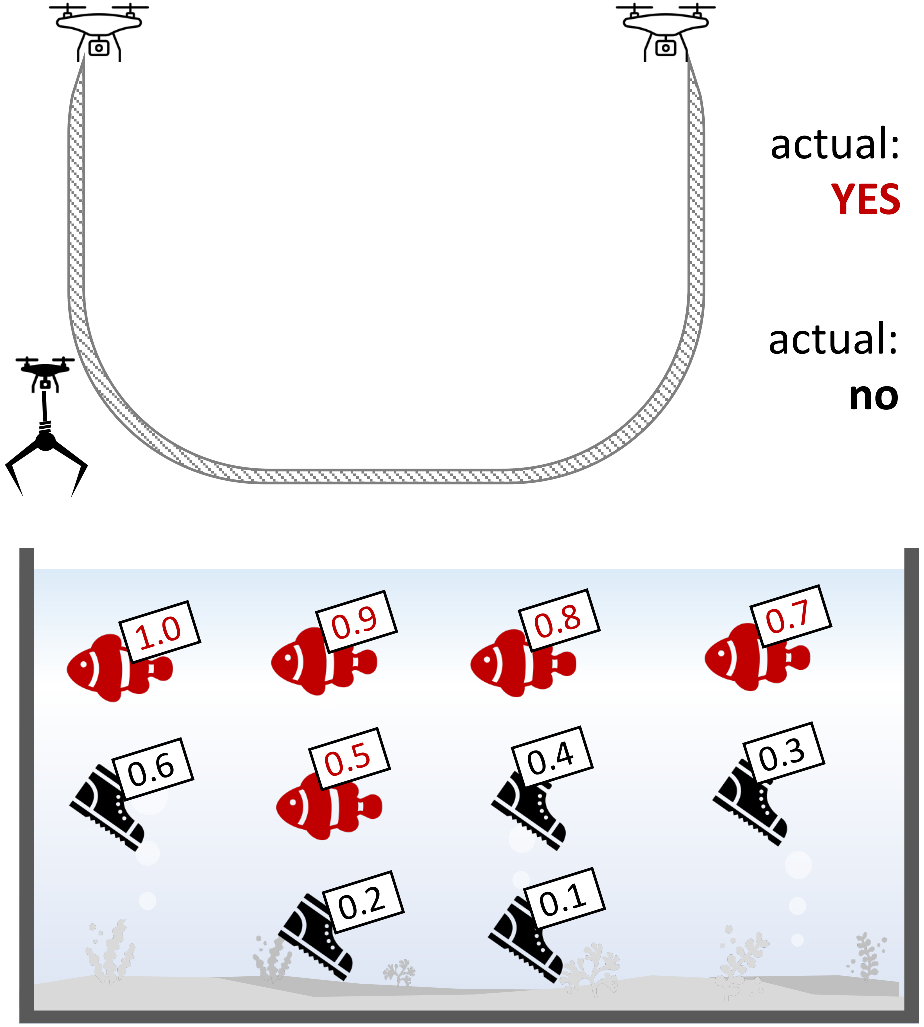


- False-Positive Rate

$$\boldsymbol{FPR} = \frac{FP}{FP + TN}$$



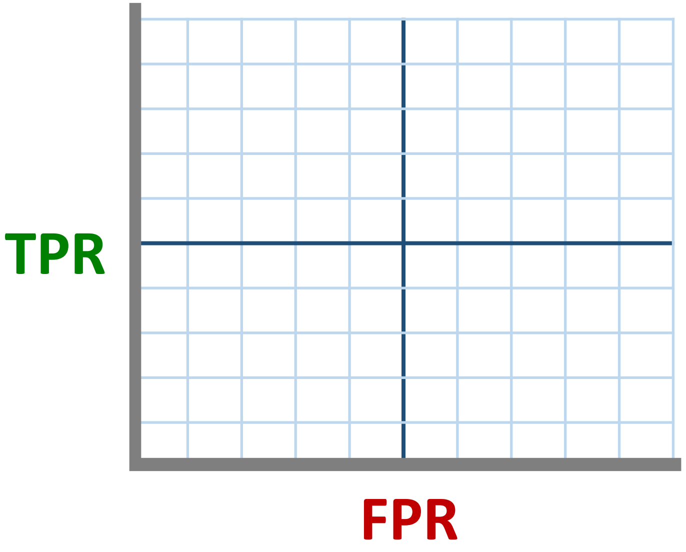
Begin



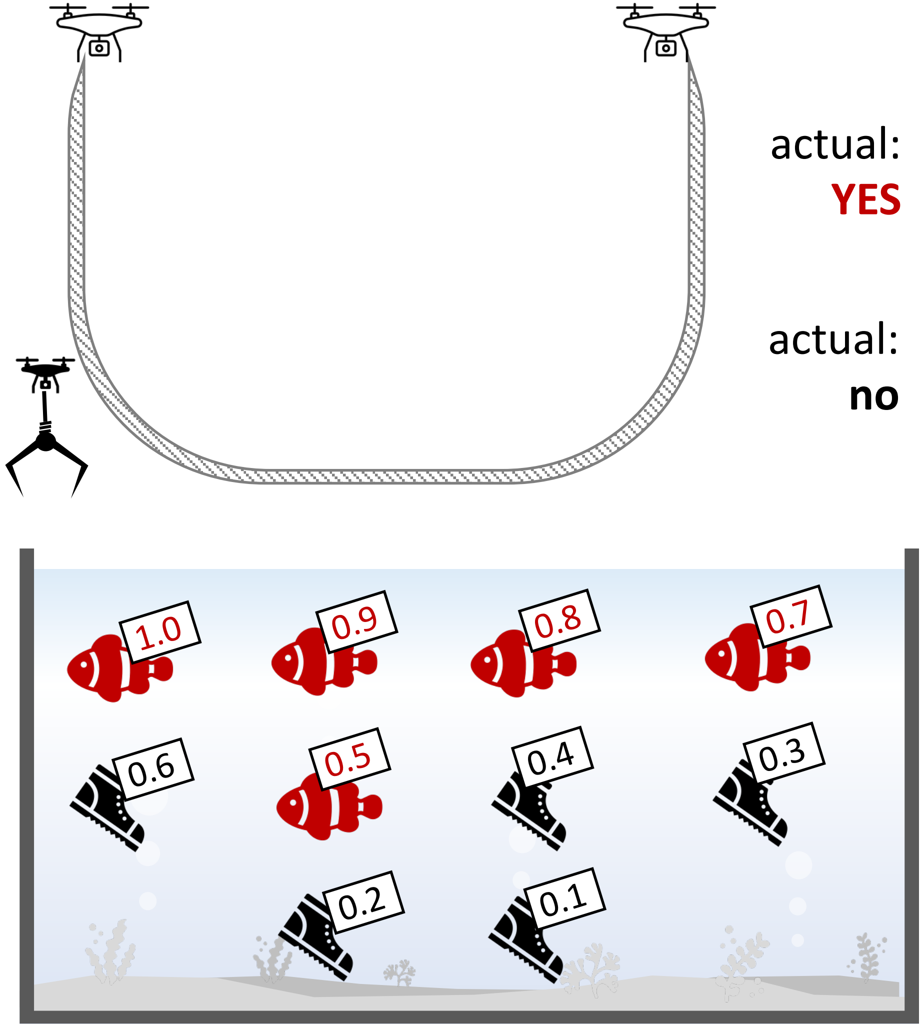
	predicted: **YES**	predicted: **no**
actual: **YES**	*(true-positive)*	*(false-negative)*
actual: **no**	*(false-positive)*	*(true-negative)*

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$



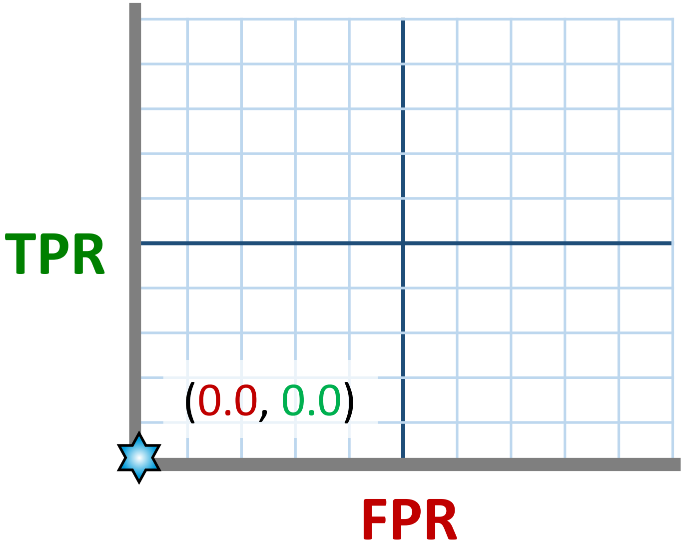
Round 0



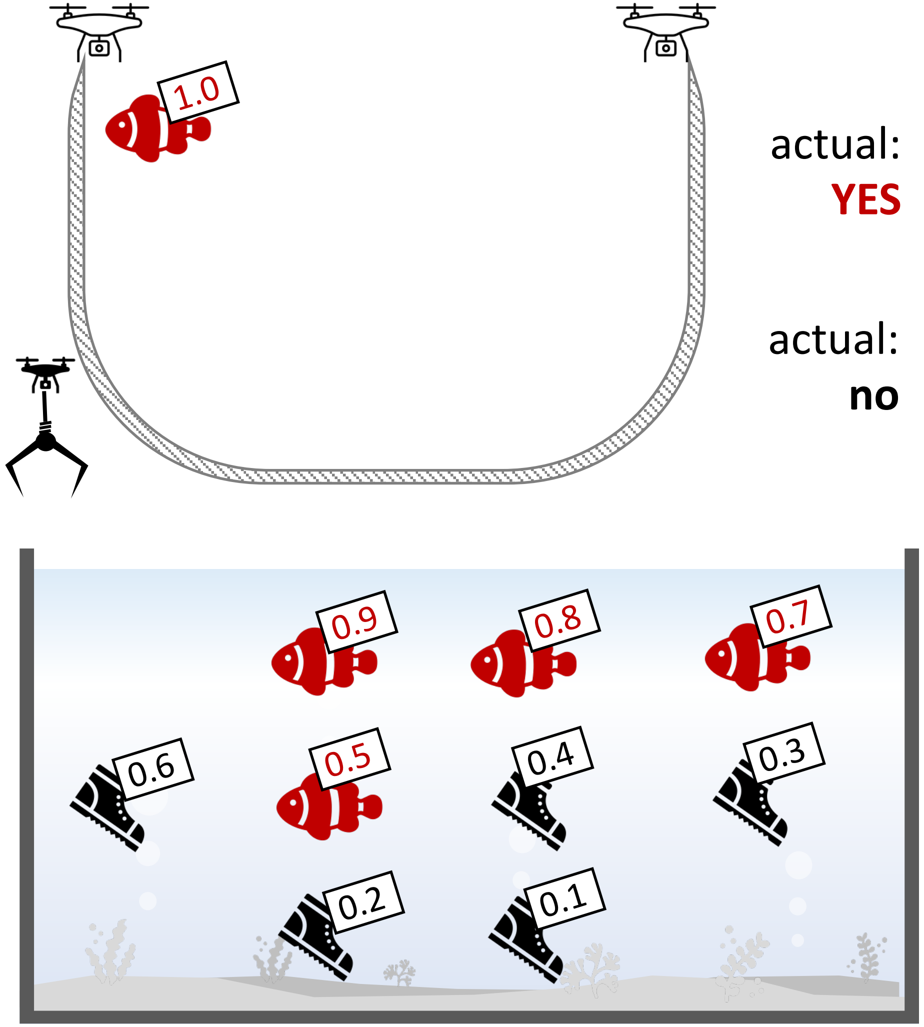
	predicted: **YES**	predicted: **no**
actual: **YES**	**0** *(true-positive)*	**5** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

$$\boldsymbol{TPR} = \frac{TP}{TP + FN} = \frac{0}{0 + 5} = 0.0$$

$$\boldsymbol{FPR} = \frac{FP}{FP + TN} = \frac{0}{0 + 5} = 0.0$$



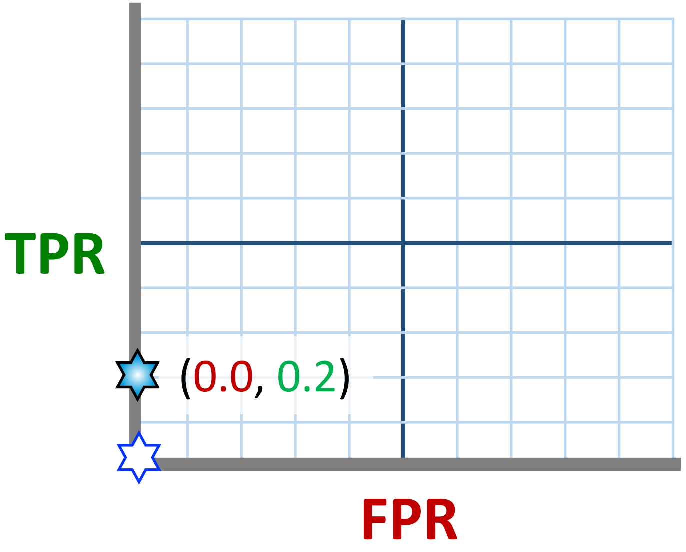
Round 1



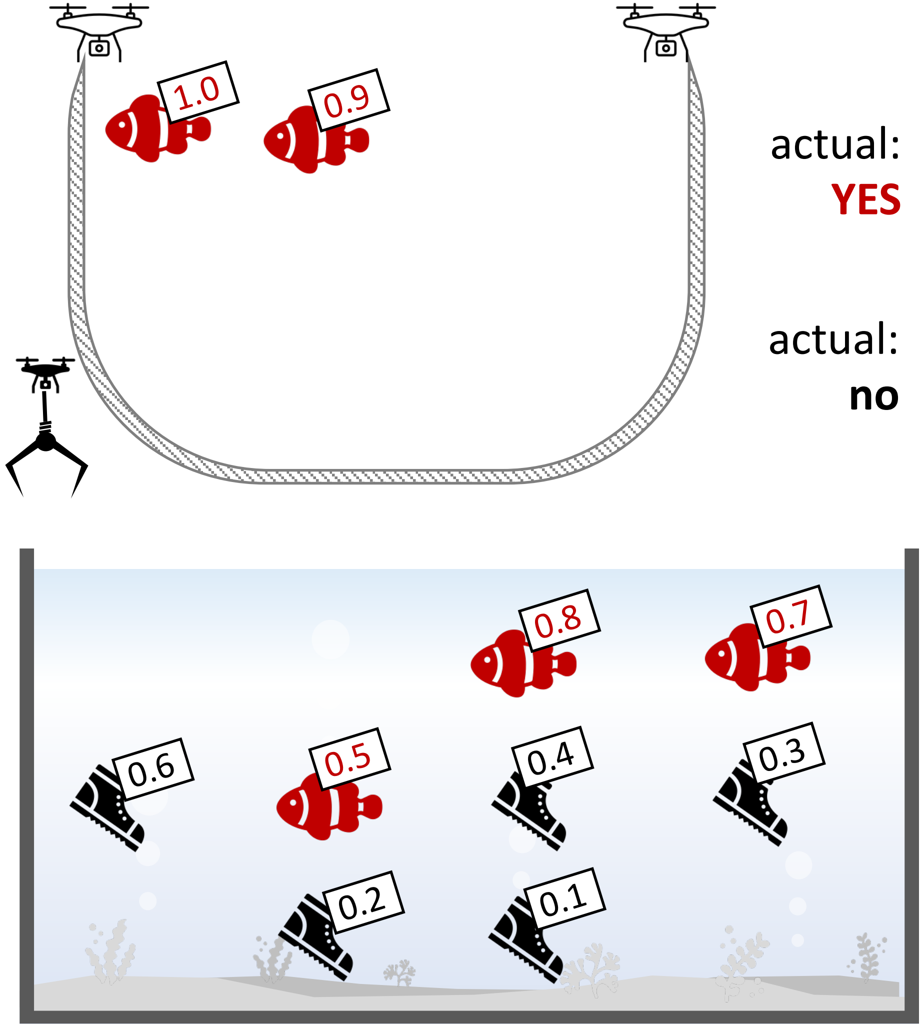
	predicted: **YES**	predicted: **no**
actual: **YES**	**1** *(true-positive)*	**4** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

$$TPR = \frac{TP}{TP + FN} = \frac{1}{1+4} = 0.2$$

$$FPR = \frac{FP}{FP + TN} = \frac{0}{0+5} = 0.0$$



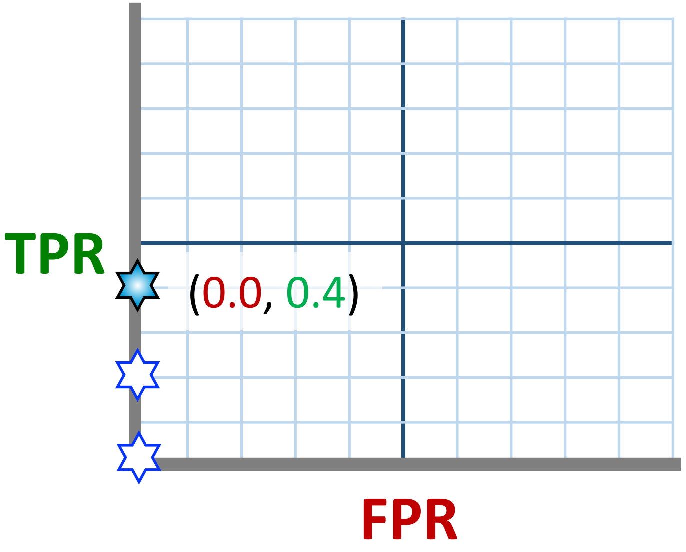
Round 2



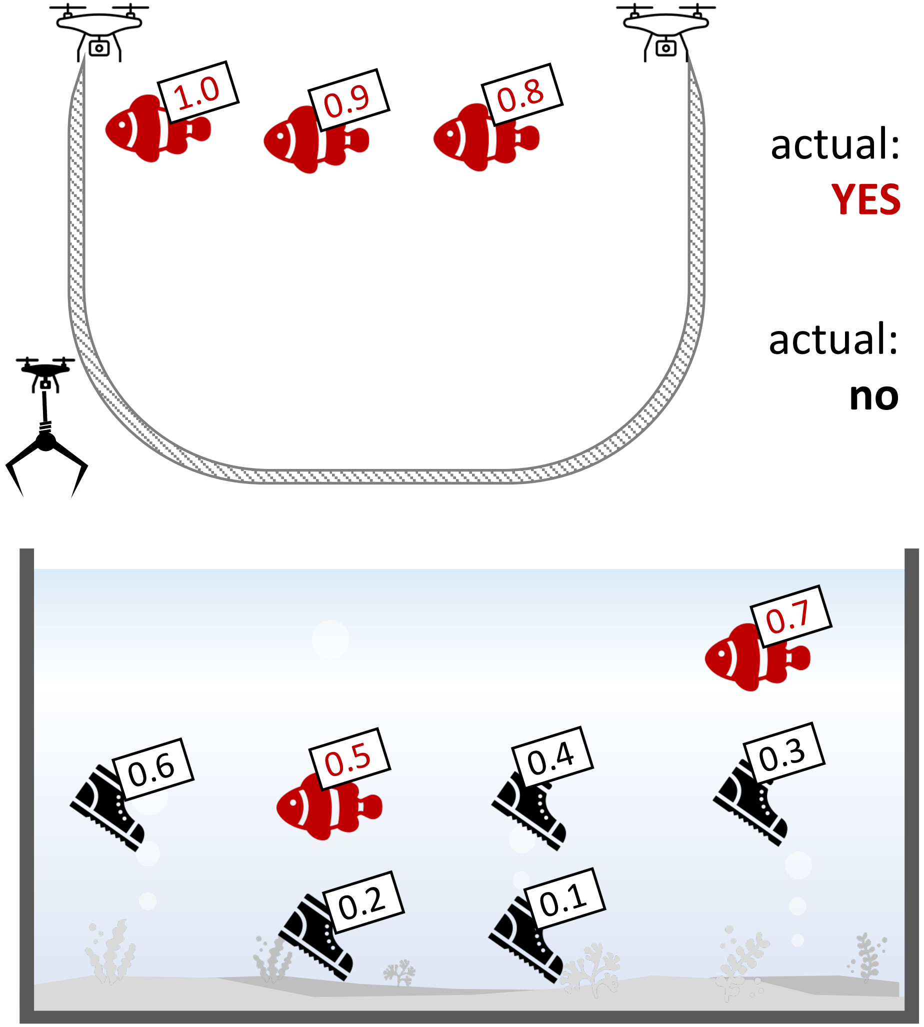
	predicted: **YES**	predicted: **no**
actual: **YES**	**2** *(true-positive)*	**3** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

$$TPR = \frac{TP}{TP + FN} = \frac{2}{2+3} = 0.4$$

$$FPR = \frac{FP}{FP + TN} = \frac{0}{0+5} = 0.0$$



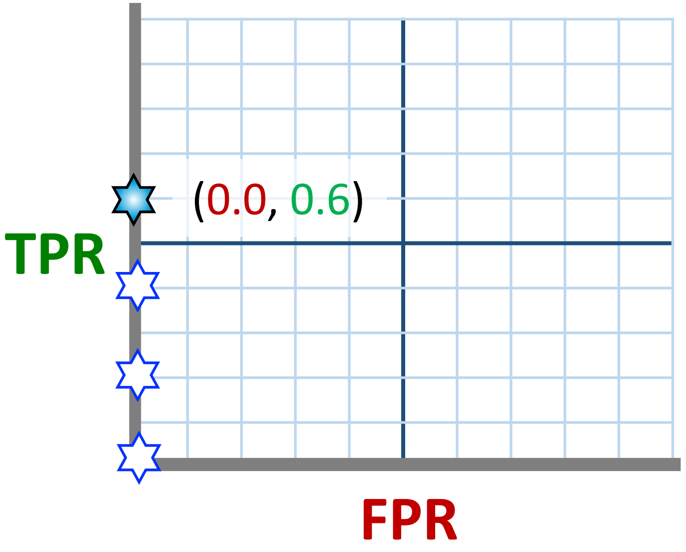
Round 3



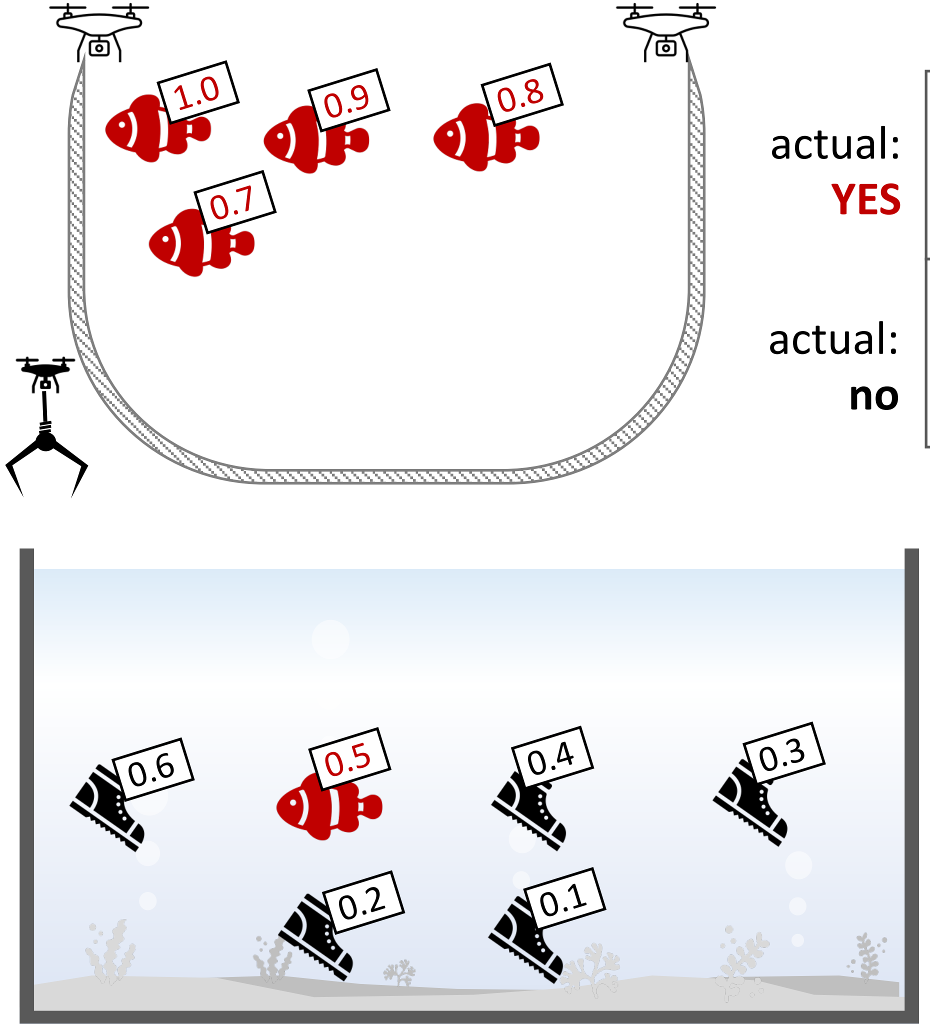
	predicted: **YES**	predicted: **no**
actual: **YES**	**3** *(true-positive)*	**2** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

$$TPR = \frac{TP}{TP + FN} = \frac{3}{3+2} = 0.6$$

$$FPR = \frac{FP}{FP + TN} = \frac{0}{0+5} = 0.0$$



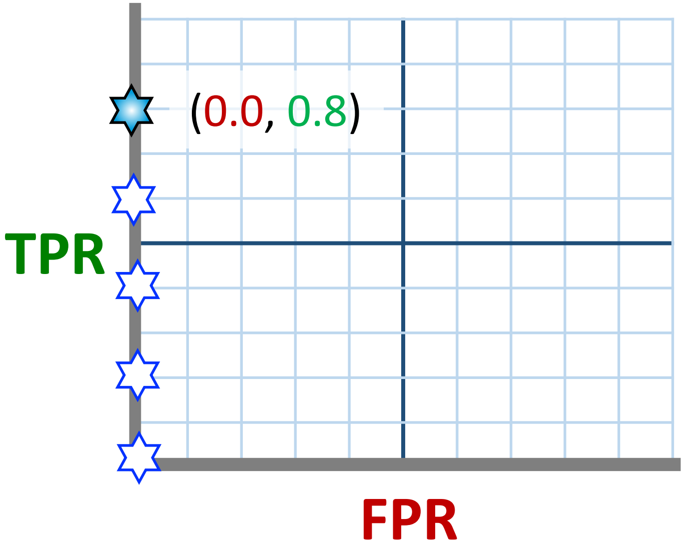
Round 4



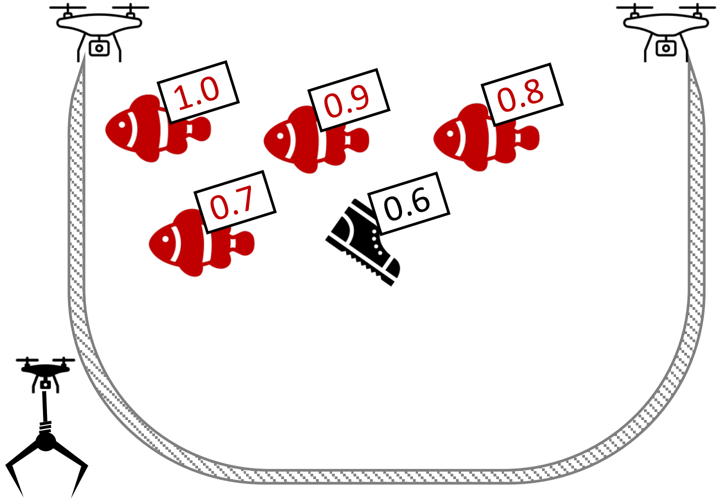
	predicted: **YES**	predicted: **no**
actual: **YES**	**4** *(true-positive)*	**1** *(false-negative)*
actual: **no**	**0** *(false-positive)*	**5** *(true-negative)*

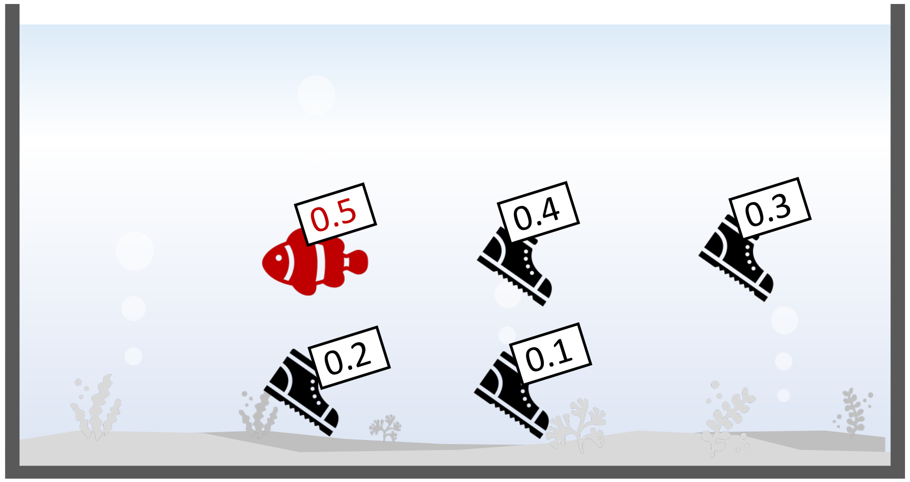
$$\boldsymbol{TPR} = \frac{TP}{TP + FN} = \frac{4}{4+1} = 0.8$$

$$\boldsymbol{FPR} = \frac{FP}{FP + TN} = \frac{0}{0+5} = 0.0$$



Round 5

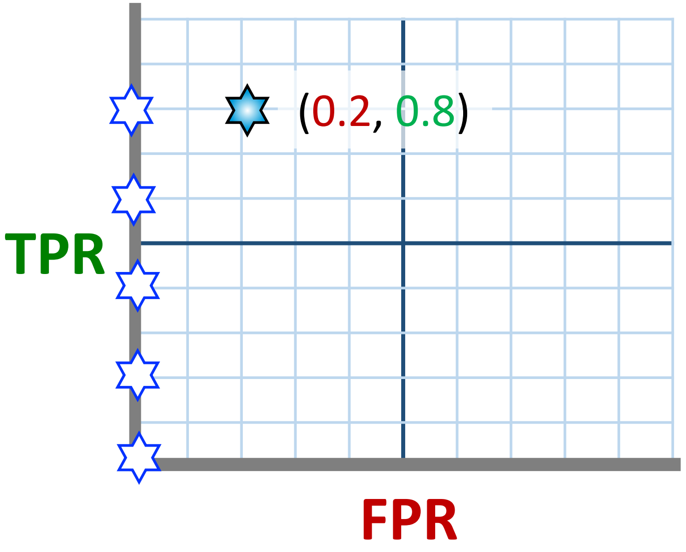




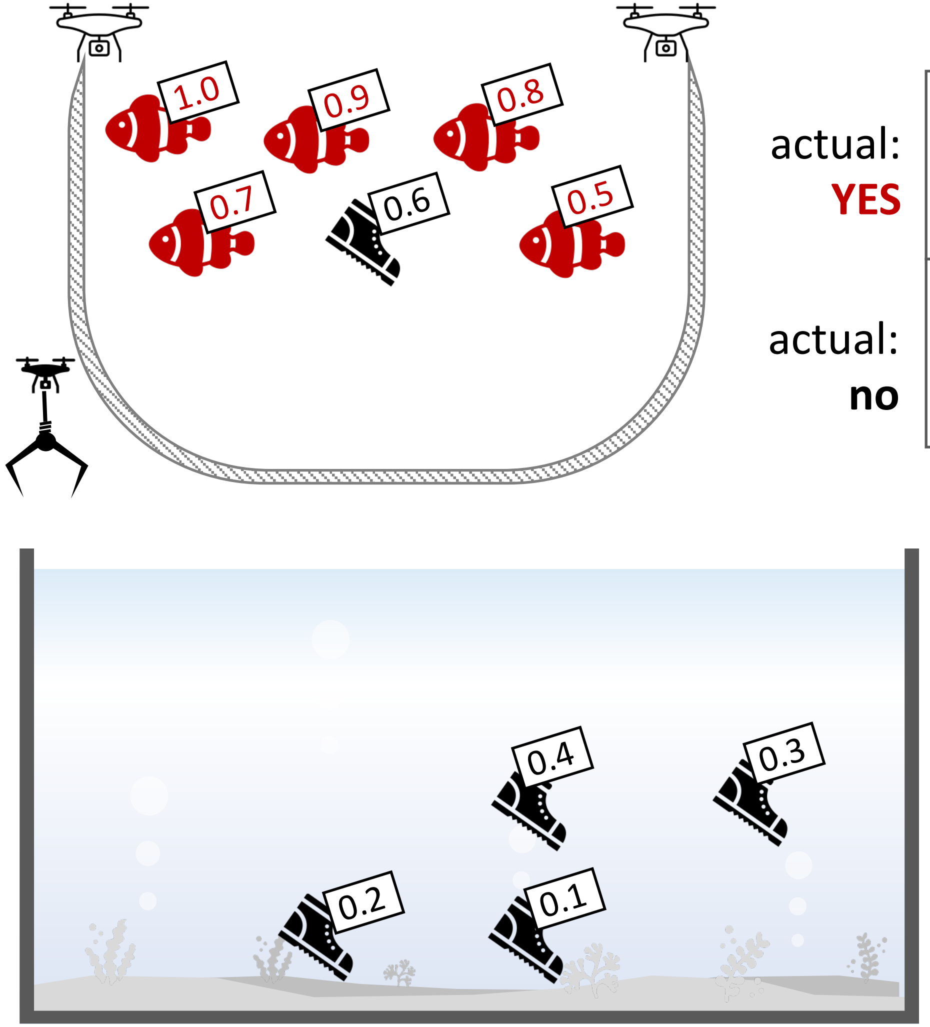
	predicted: **YES**	predicted: **no**
actual: **YES**	**4** *(true-positive)*	**1** *(false-negative)*
actual: **no**	**1** *(false-positive)*	**4** *(true-negative)*

$$TPR = \frac{TP}{TP + FN} = \frac{4}{4 + 1} = 0.8$$

$$FPR = \frac{FP}{FP + TN} = \frac{1}{1 + 4} = 0.2$$



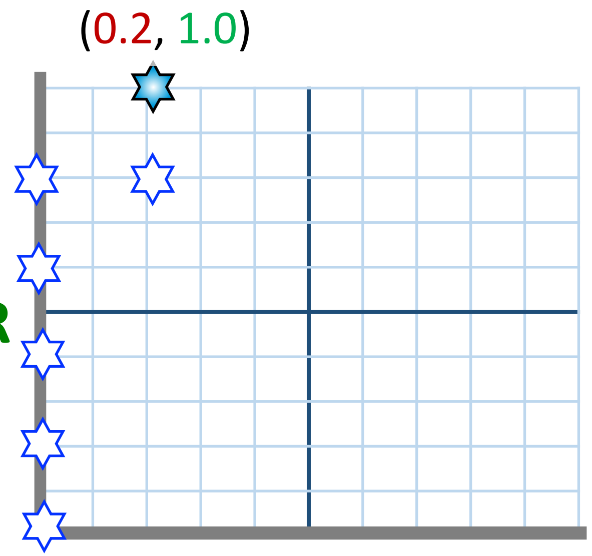
Round 6



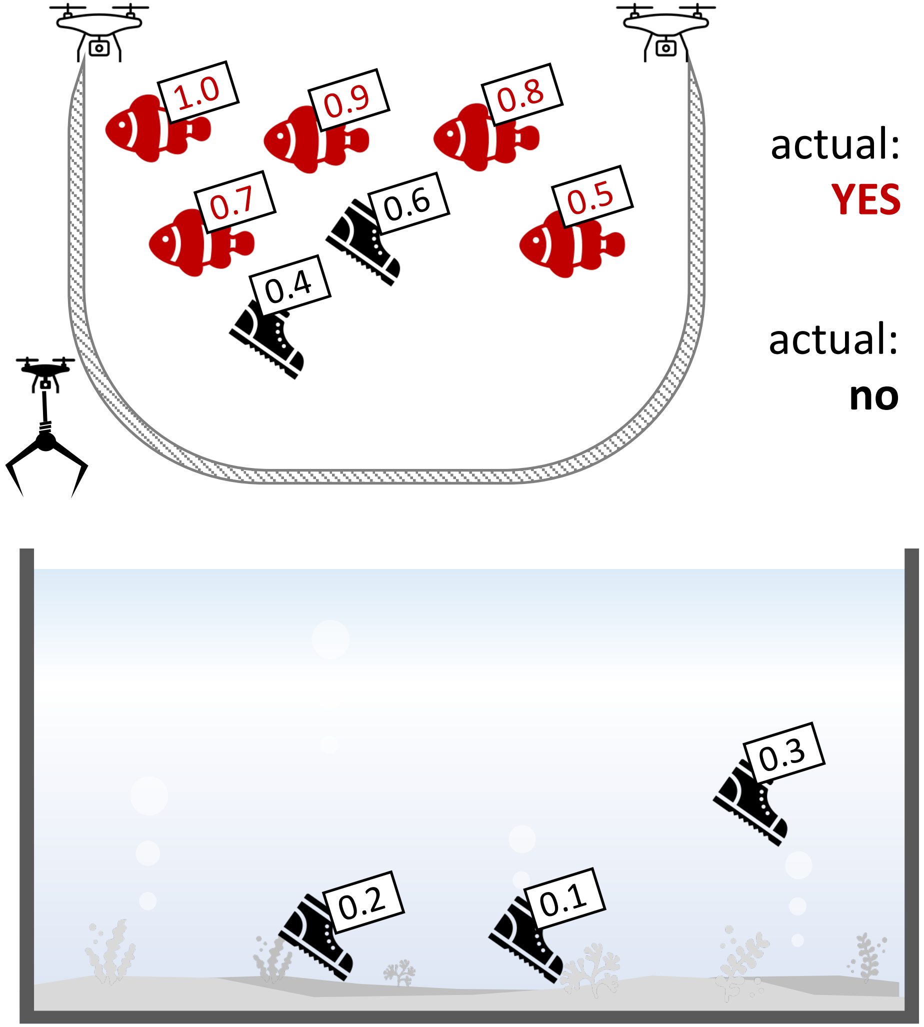
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** (true-positive)	**0** (false-negative)
actual: **no**	**1** (false-positive)	**4** (true-negative)

$$TPR = \frac{TP}{TP + FN} = \frac{5}{5+0} = 1.0$$

$$FPR = \frac{FP}{FP + TN} = \frac{1}{1+4} = 0.2$$



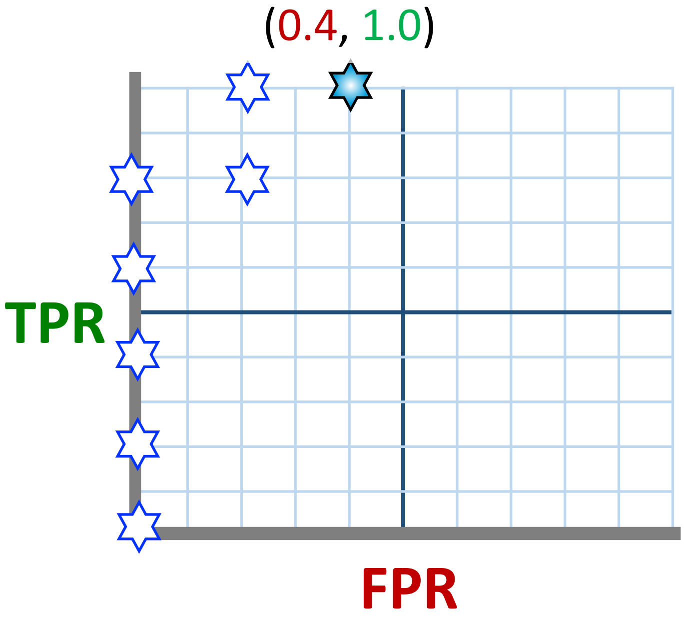
Round 7



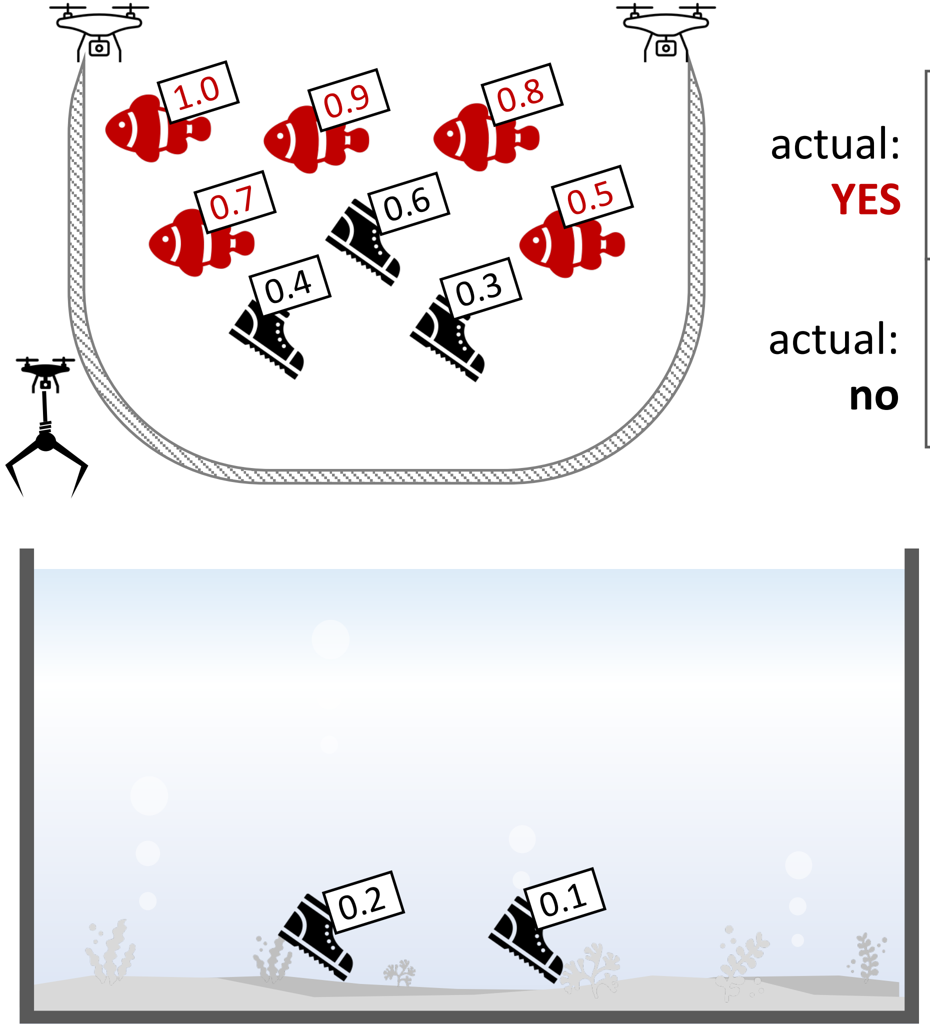
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**2** *(false-positive)*	**3** *(true-negative)*

$$TPR = \frac{TP}{TP + FN} = \frac{5}{5 + 0} = 1.0$$

$$FPR = \frac{FP}{FP + TN} = \frac{2}{2 + 3} = 0.4$$



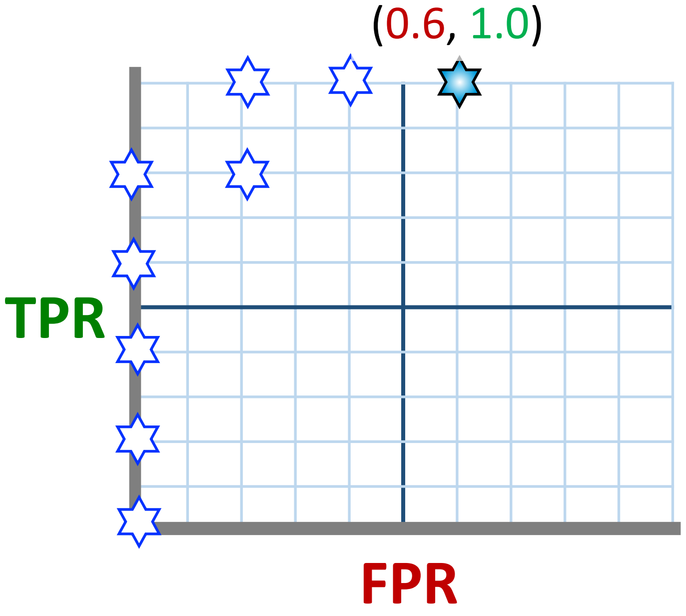
Round 8



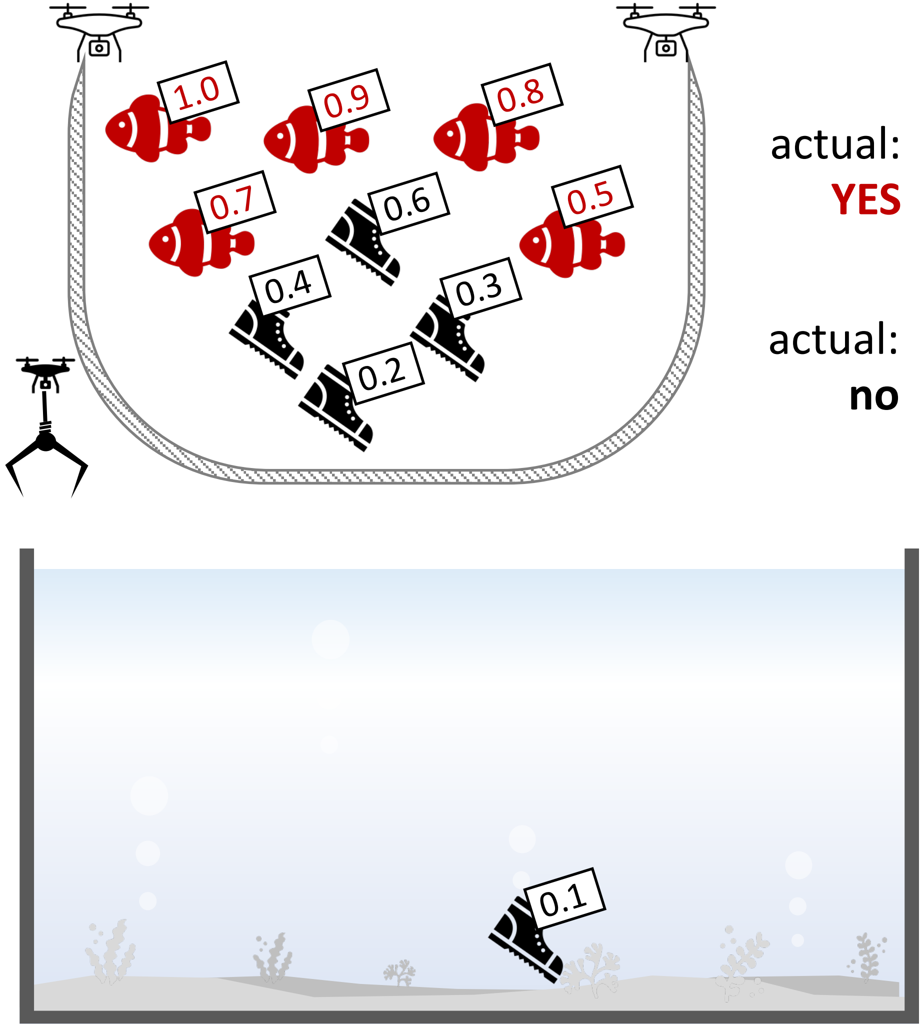
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**3** *(false-positive)*	**2** *(true-negative)*

$$\boldsymbol{TPR} = \frac{TP}{TP + FN} = \frac{5}{5 + 0} = 1.0$$

$$\boldsymbol{FPR} = \frac{FP}{FP + TN} = \frac{3}{3 + 2} = 0.6$$



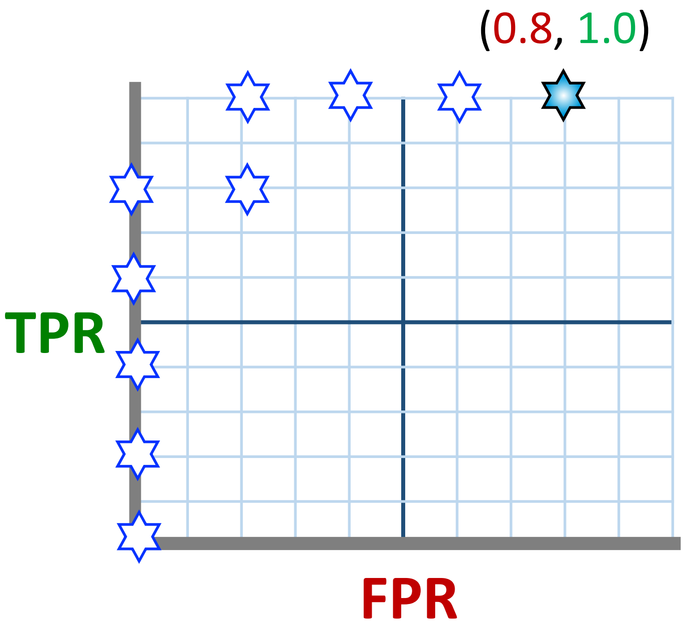
Round 9



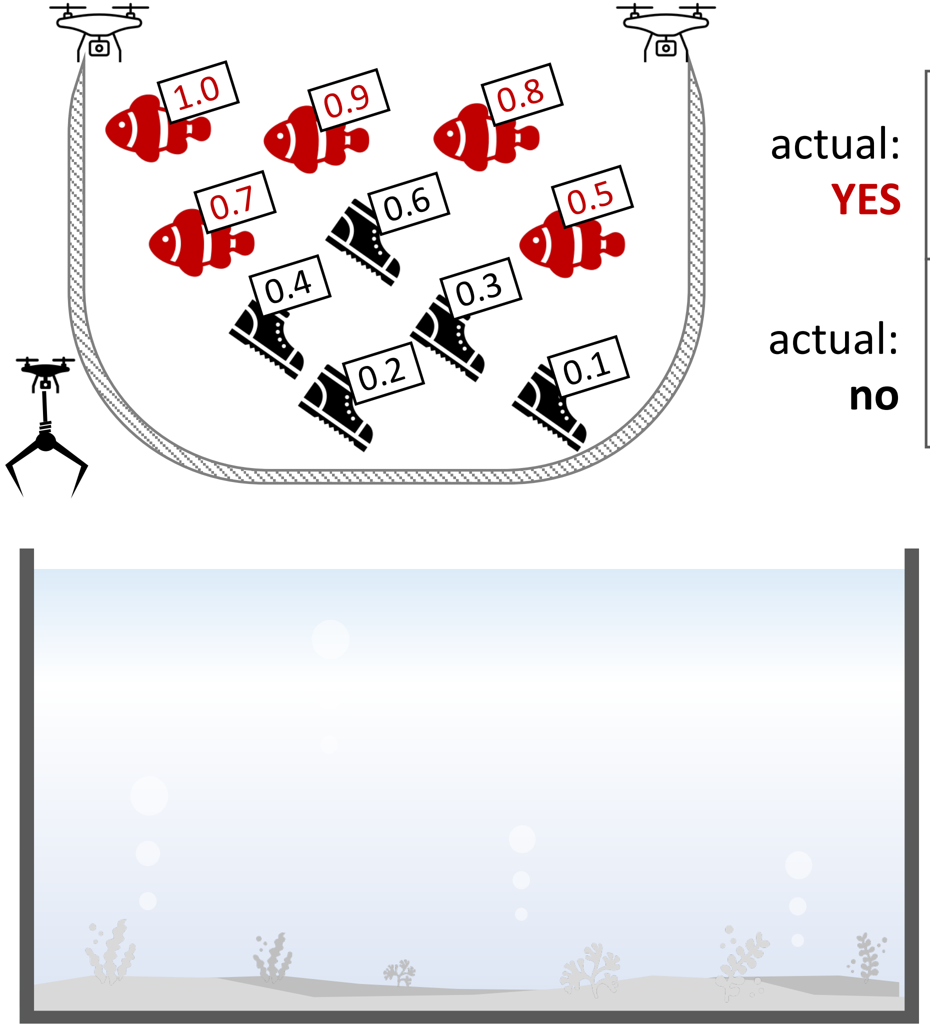
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**4** *(false-positive)*	**1** *(true-negative)*

$$TPR = \frac{TP}{TP + FN} = \frac{5}{5+0} = 1.0$$

$$FPR = \frac{FP}{FP + TN} = \frac{4}{4+1} = 0.8$$



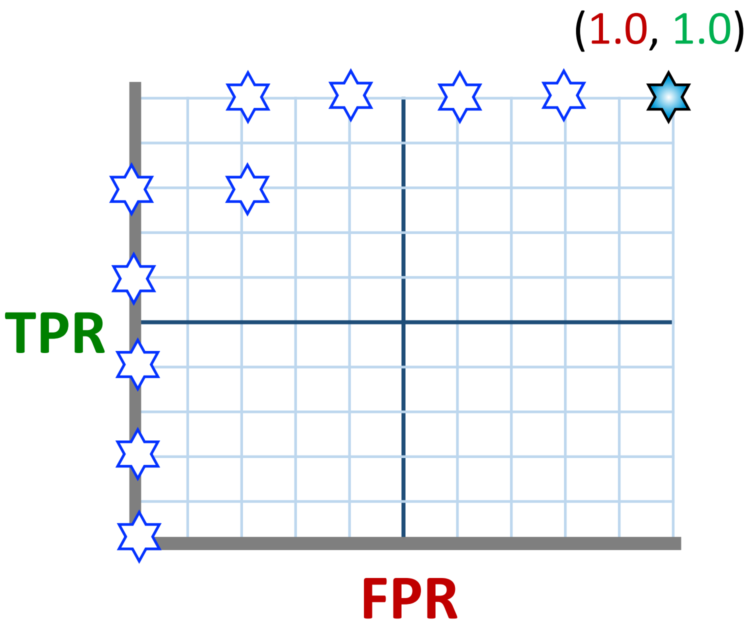
Round 10



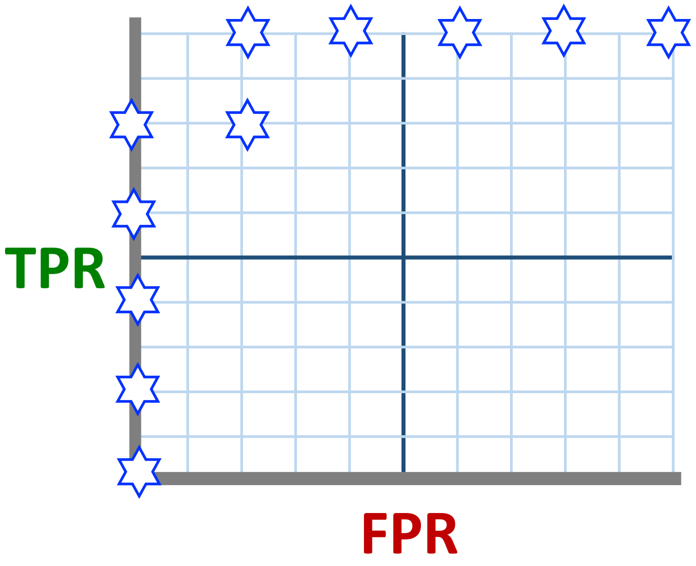
	predicted: **YES**	predicted: **no**
actual: **YES**	**5** *(true-positive)*	**0** *(false-negative)*
actual: **no**	**5** *(false-positive)*	**0** *(true-negative)*

$$\boldsymbol{TPR} = \frac{TP}{TP + FN} = \frac{5}{5+0} = 1.0$$

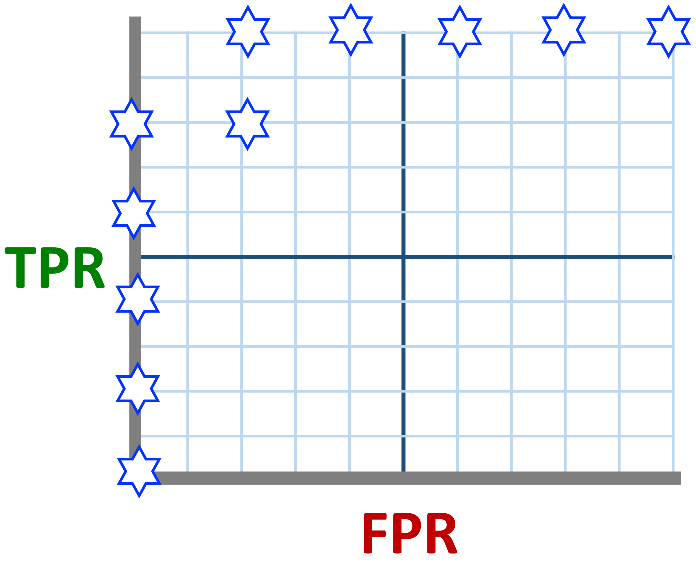
$$\boldsymbol{FPR} = \frac{FP}{FP + TN} = \frac{5}{5+0} = 1.0$$



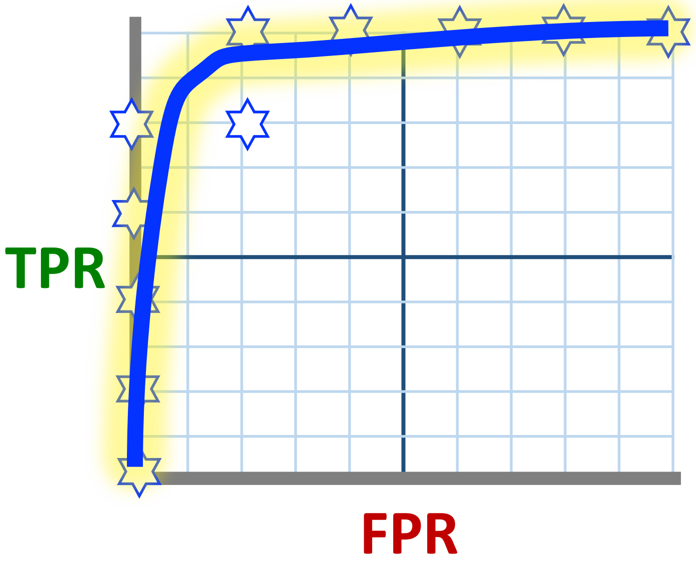
TPR & FPR



TPR & FPR

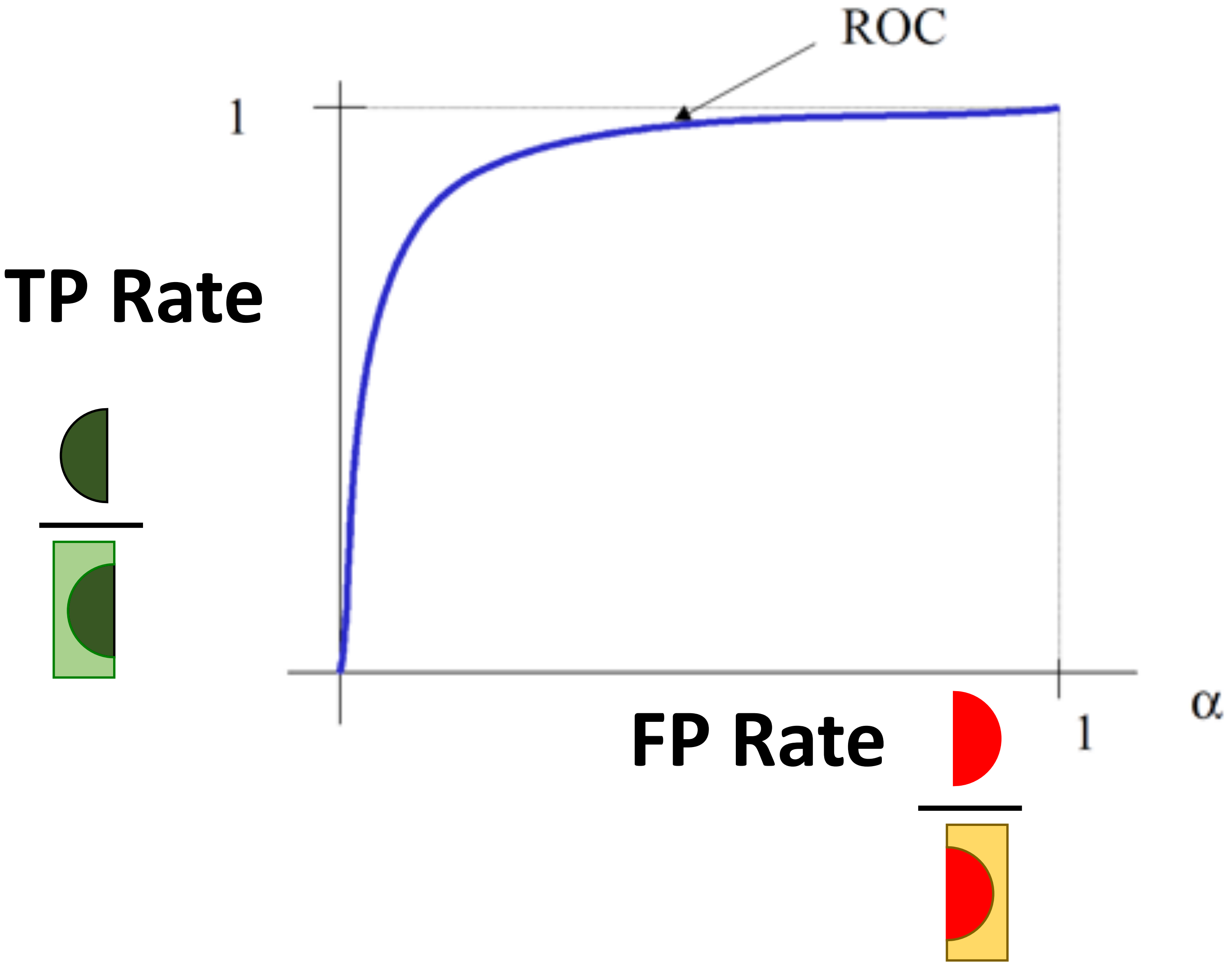


ROC Curve



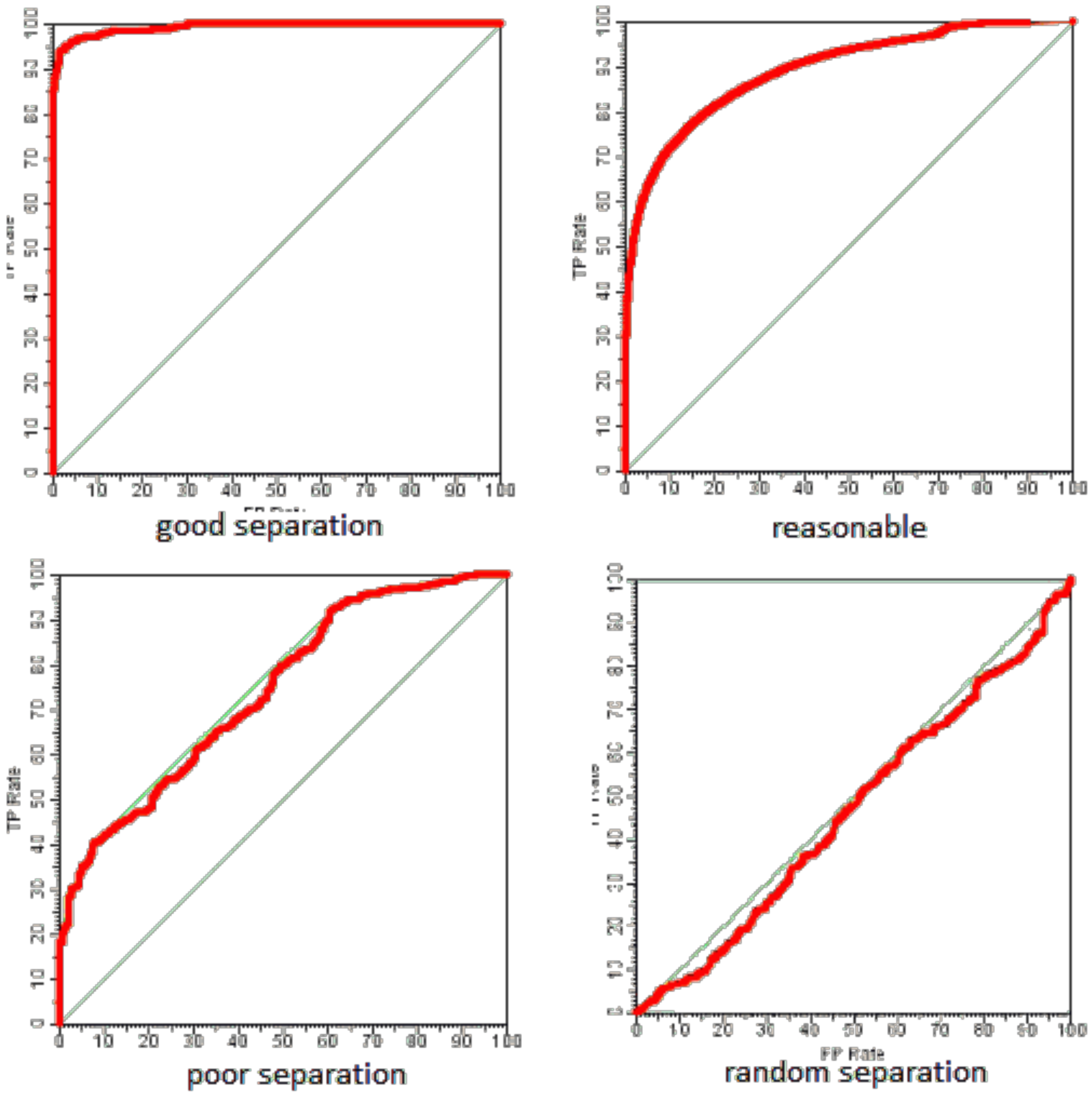
ROC Curve

- Receive Operating Characteristic



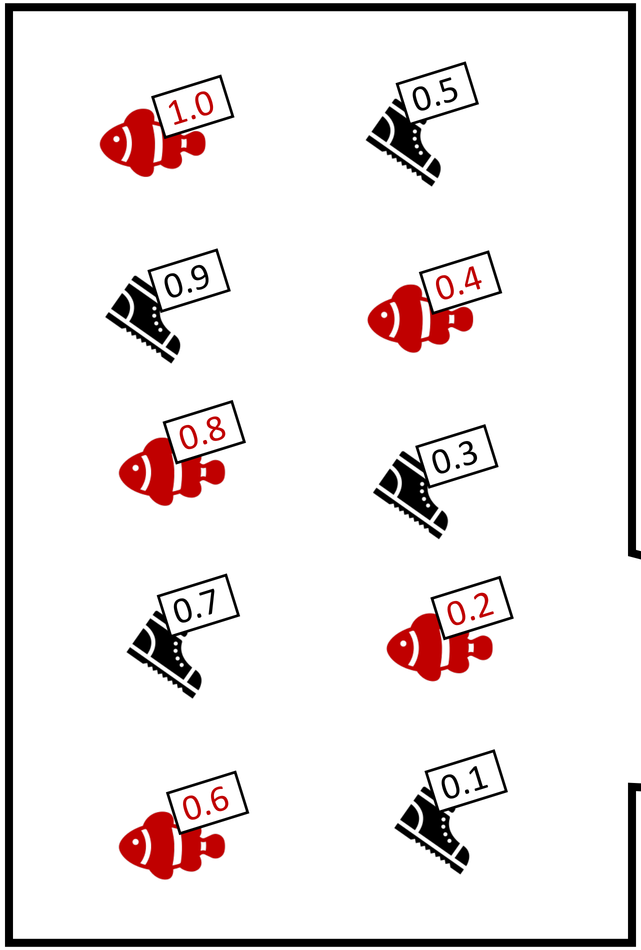
Ref: https://www.projectrhea.org/rhea/index.php/NeymanPearson_Lemma_and_Receiver_Operating_Characteristic_Curve

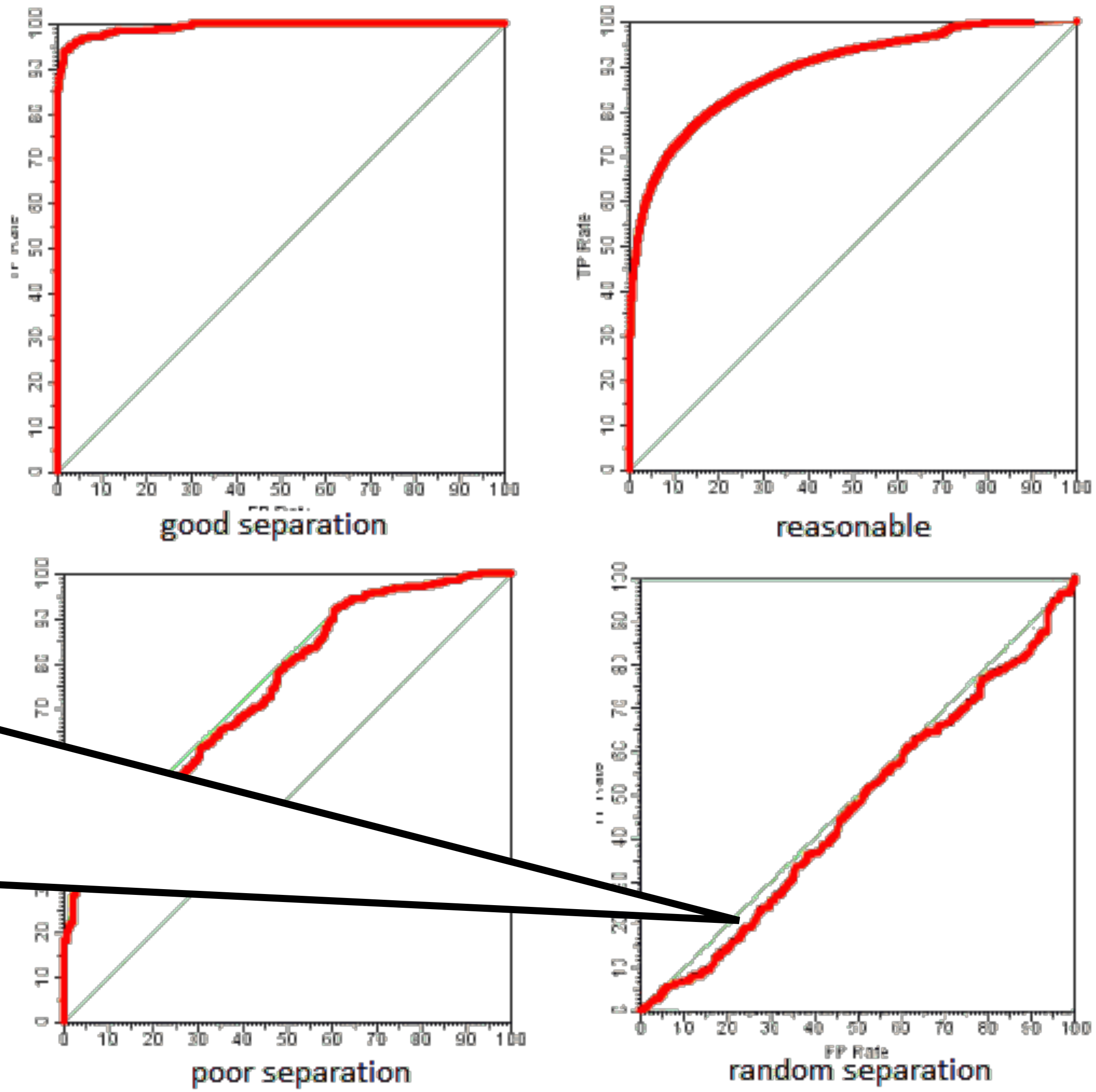
ROC Curve



Ref: http://mlwiki.org/index.php/ROC_Analysis

ROC Curve





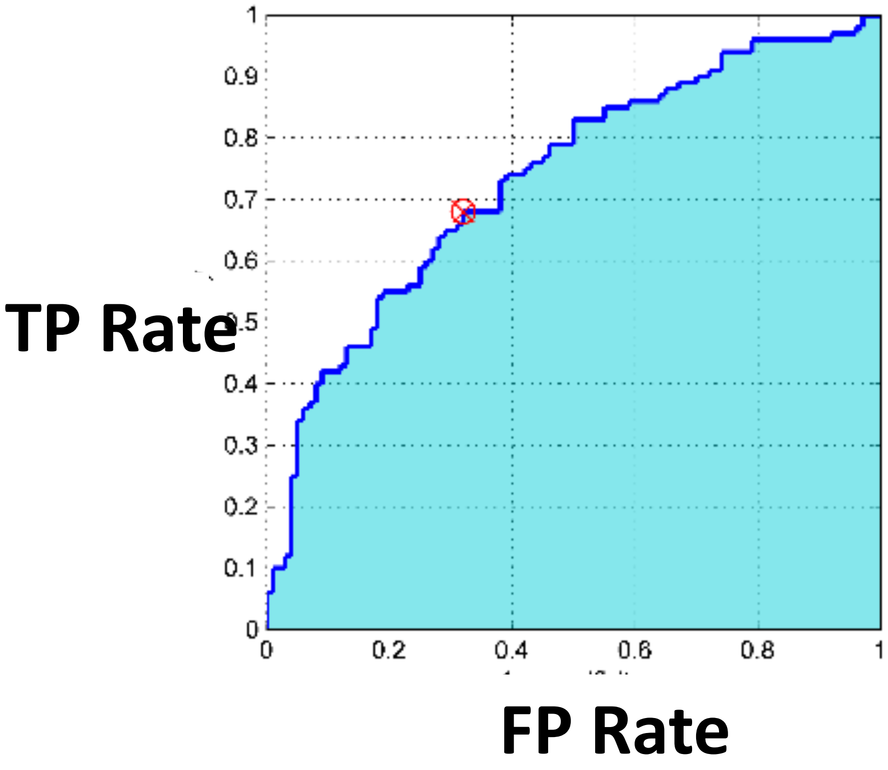
Ref: http://mlwiki.org/index.php/ROC_Analysis

AUC : Area Under the ROC Curve

The graph at right shows three ROC curves representing excellent, good, and worthless tests plotted on the same graph. An area of 1 represents a perfect test; an area of 0.5 represents a worthless test. A rough guide for classifying the accuracy of a diagnostic test is the traditional academic point system:

- 0.9 - 1.0 = excellent (A)
- 0.8 - 0.9 = good (B)
- 0.7 - 0.8 = fair (C)
- 0.6 - 0.7 = poor (D)
- 0.5 - 0.6 = fail (F)

AUC = 0.7



Ref: http://gim.unmc.edu/dxtests/roc3.htm
http://molevol.altervista.org/blog/roc-curve-area-roc-curve-auc/

Confusion Matrix for Multi-Class

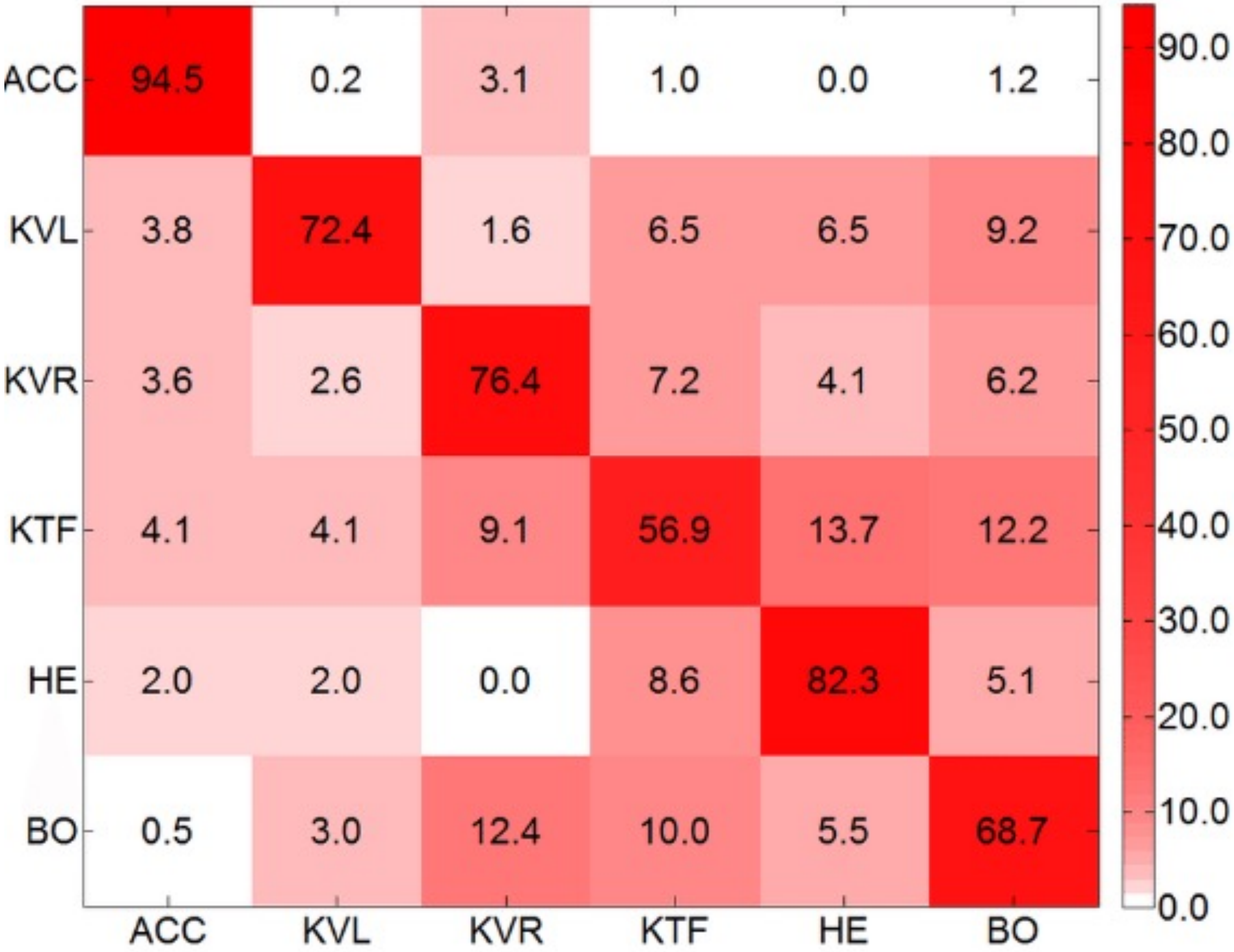
Confusion Matrix for Multi-Class

Predicted Class (columns) / **Actual Class** (rows)

	Cat	Dog	Rabbit
Cat	5	2	0
Dog	3	3	2
Rabbit	0	1	11

Ref: http://python-data-science.readthedocs.io/en/latest/evaluation.html

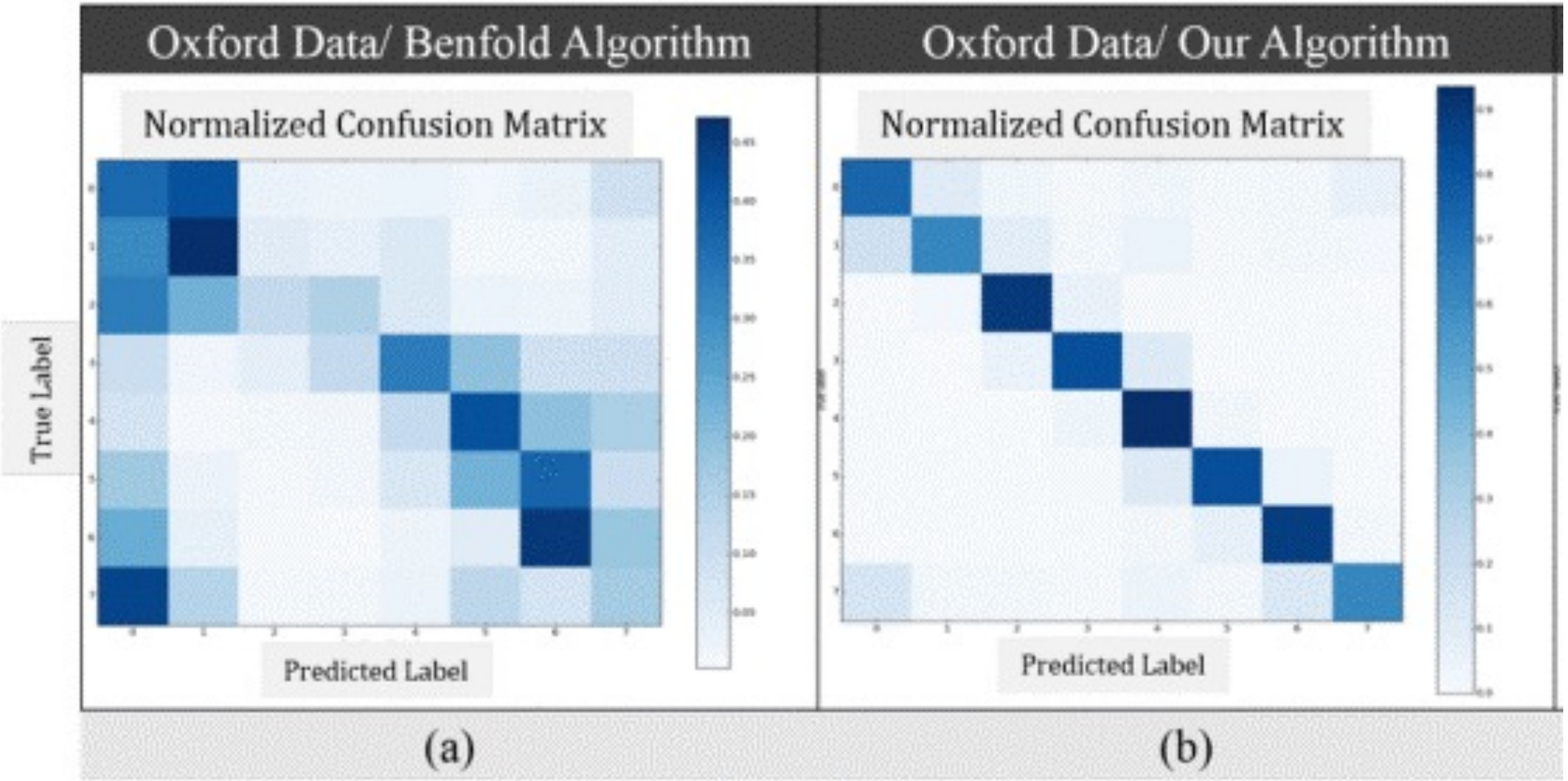
Heatmap



Ref: https://www.researchgate.net/figure/Figure-3-Heat-map-confusion-matrix-for-multi-label-exact-deviation-classification-results-using_316316283_fig3

Comparison

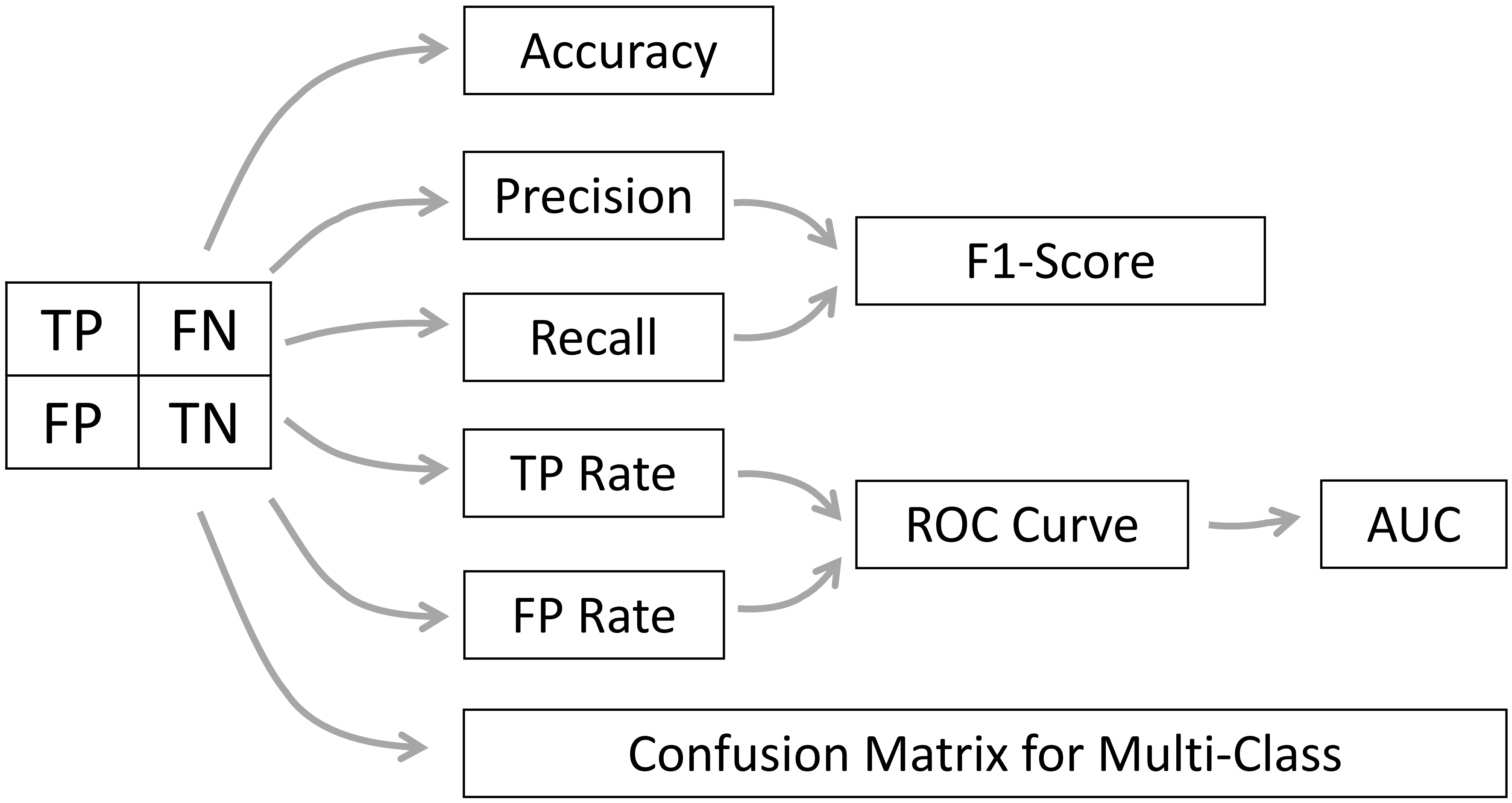
- Which one is better?



Ref: http://ieeexplore.ieee.org/document/7279167/

Summary

Evaluation Methods



Recommendation

- Balanced Binary-Class Distribution
  - Accuracy
- Imbalanced Binary-Class Distribution
  - Precision, Recall, F1
- Multi-Class
  - Precision, Recall, F1
- Overall Peformance of the Algorithm
  - ROC Curve, AUC
- For Transparency Result
  - show the confusion Matrix